

Deep Tangency Portfolio *

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Abstract

We develop a deep learning framework to directly estimate tangency portfolio weights by augmenting benchmark factors with a long-short deep factor derived from high-dimensional characteristics. Guided by a Sharpe ratio maximization objective, the deep factor plays two key roles: (i) providing a natural hedge against benchmark risks, and (ii) capturing multi-market signals to enhance the performance of the benchmark. In an empirical application to U.S. corporate bonds using 132 firm-level bond, equity, and option characteristics, the deep factor achieves an out-of-sample Sharpe ratio exceeding 1, and the deep tangency portfolio benchmarked against the market factor yields a Sharpe ratio over 2, outperforming portfolios spanned by common observable or latent factors. These results demonstrate that bond risk signals are fundamentally dense, requiring the nonlinear integration of multi-market information to span the efficient frontier.

Keywords: Tangency Portfolio, Deep Learning, Deep Factor, Enhancement Device, Corporate Bonds

JEL Classification: C45, G11, G12.

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1 Introduction

The modern portfolio theory of [Markowitz \(1952\)](#) defines the mean-variance efficient (MVE) portfolio as $\Sigma^{-1}\mu$, where Σ is the covariance matrix of asset returns and μ is the vector of expected returns. However, as [Cochrane \(2014, p.7\)](#) emphasizes, this solution becomes impractical in high-dimensional settings due to the significant estimation challenges inherent in both inputs. To alleviate these issues, the literature often relies on a small number of characteristic-managed factors (e.g., [Fama and French, 1996, 2015](#)). But these sparse, linear factor models fail to span the efficient frontier because return signals are fundamentally dense across a high-dimensional characteristic set (e.g., [Kozak, Nagel, and Santosh, 2018](#); [Avramov, Feng, He, and Xiao, 2025](#); [Dickerson, Julliard, and Mueller, 2026](#)).

We develop a deep learning framework designed as an investment *enhancement device* to improve the performance of well-established strategies, such as the market portfolio or multi-factor benchmarks. By formalizing benchmark factors as an economic prior, our framework imposes regularization on the neural network’s estimation, thereby mitigating the challenge of low signal-to-noise ratios in financial data. This design allows the model to utilize nonlinear transformations to extract residual, multi-market signals that benchmark factors overlook. In addition, rather than separately estimating expected returns and the covariance matrix of individual assets, we directly estimate portfolio weights by maximizing the Sharpe ratio.¹ As such, our framework sidesteps the compounded estimation errors and noise sensitivity inherent in traditional plug-in methods ([Brandt, Santa-Clara, and Valkanov, 2009](#)), thereby translating statistical predictive power into tangible utility gains for a mean-variance investor.

Specifically, our deep learning architecture endogenously compresses high-dimensional

¹The recent literature with similar economic goals but in different frameworks includes [Guijarro-Ordóñez, Pelger, and Zanotti \(2025\)](#), [Bryzgalova, Pelger, and Zhu \(2025\)](#), among others.

characteristics into a single latent characteristic per asset, forming the basis for constructing our long-short deep factor. While rooted in the familiar characteristic-sorting scheme, this construction enables flexible, nonlinear transformations scalable to large datasets. The resulting deep factor thus systematically complements benchmark strategies by uncovering omitted signals, offering two primary benefits: (i) endogenously acting as a potential hedge against benchmark risks driven by the Sharpe ratio maximization; and (ii) expanding the efficient frontier by capturing signals overlooked by the benchmark factors. Ultimately, our framework connects parsimonious factor investing with the optimal portfolio in a high-dimensional asset universe (Kozak, Nagel, and Santosh, 2020).

Our framework is highly adaptable: it can integrate alternative benchmark factors, accommodate diverse objectives such as variance minimization, and regularize extreme portfolio weights (see, e.g., DeMiguel, Garlappi, Nogales, and Uppal, 2009; Avramov, Cheng, and Metzker, 2023). While applicable across various settings, these advantages are most pronounced in markets where traditional factor models fail to explain cross-sectional return spreads. The U.S. corporate bond market represents a primary case; pricing here remains less transparent than in equities, and standard factors often fail to span the efficient frontier — a challenge well-documented in equities (Kozak et al., 2018; Cong, Feng, He, and He, 2025) and increasingly evident in bonds (Dickerson, Mueller, and Robotti, 2023; Dickerson et al., 2026). Moving beyond the constraints of traditional term and default factors (Fama and French, 1993) or the CAPM (Dickerson et al., 2023), our approach recovers the high-dimensional information set that sparse models typically overlook.² Although Elkamhi, Jo, and Nozawa (2024) show that a one-factor model based on long-run consumption growth prices bond portfolios, our results show that nonlinearly integrating a rich set of character-

²Other bond factors include liquidity (Lin, Wang, and Wu, 2011), momentum (Jostova, Nikolova, Philipov, and Stahel, 2013), volatility (Chung, Wang, and Wu, 2019), and long-term reversal (Bali, Subrahmanyam, and Wen, 2021).

istics is needed to fully span the efficient frontier. [Kozak and Nagel \(2024\)](#) show that characteristic-managed factors hardly span the efficient frontier unless a large number of characteristics are used simultaneously.

Applying our approach to the corporate bond market, we evaluate the deep tangency portfolio using monthly corporate bond returns from the enhanced Trade Reporting and Compliance Engine (TRACE). We use 132 firm-level characteristics from three sources: (i) 41 bond-specific variables from the merged TRACE-FISD data; (ii) 61 equity characteristics that capture the fundamental links between bond and stock prices; and (iii) 30 option-related variables. The inclusion of option variables is particularly novel in this context, motivated by their predictive power for bond returns ([Chung et al., 2019](#); [Huang, Jiang, and Li, 2023](#); [Cao, Goyal, Xiao, and Zhan, 2023](#)). By synthesizing these multi-market signals, particularly the underutilized option-implied data, our framework aims to extract predictive information from the firm’s entire capital structure.

Our main analysis integrates the full characteristic set with the equal- or value-weighted bond market factor as the benchmark for the period from July 2004 to December 2020. We allocate the period from July 2004 to June 2014 for model training and validation, and the period from July 2014 to December 2020 for out-of-sample (OOS) evaluation. For comparison, we also construct alternative optimal portfolios spanned by the Fama-French (FF) five factors, including three bond factors (market, term, and default factors) and two equity factors (size and value factors) ([Fama and French, 1993, 1996](#)), the five latent factors derived from the instrumental principal component analysis (IPCA) ([Kelly, Pruitt, and Su, 2019](#)), and the five latent factors derived from the risk-premium principal component analysis (RP-PCA) ([Lettau and Pelger, 2020](#)).

The key findings can be summarized as follows. First, we find that in our framework, a 2-layer neural network performs the best. It is important to impose a nonlinear long-short structure: a deep factor with linear ranking underperforms one with soft-

max ranking, which explicitly enforces this structure. The resulting softmax-ranking-based deep factor achieves out-of-sample annualized Sharpe ratios of 1.0 and 1.49 when the equal- and value-weighted bond market factors are used as benchmarks, respectively. Due to the Sharpe ratio maximization objective, the deep factor exhibits a negative correlation with the benchmark, and both rarely decline simultaneously, particularly during market downturns, making it an effective hedging portfolio. Consequently, the deep tangency portfolio that combines the deep factor and the equal-weighted (value-weighted) benchmark in our deep learning framework achieves an OOS annualized Sharpe ratio of 2.13 (2.21), substantially enhancing risk-adjusted returns compared to the benchmark.

Second, our deep tangency portfolio outperforms alternative optimal portfolios constructed from commonly used observable and latent factors. For example, the optimal portfolio spanned by the Fama-French five observable factors generates an OOS annualized Sharpe ratio of only 0.88, and the optimal portfolios spanned by IPCA and RP-PCA five latent factors achieve slightly higher OOS annualized Sharpe ratios. The substantial improvement in performance when augmenting traditional portfolios with our deep factor suggests that the enhancement gains are not mere artifacts of data mining, but instead represent the spanning of a residual efficient frontier, capturing a dense network of return signals that traditional parsimonious models fundamentally overlook. Factor spanning regressions further show that these observable and latent factors are insufficient to explain the performance of the deep factor and the deep tangency portfolio. We further show that explicitly incorporating benchmark factors into our deep learning framework expands the efficient frontier, evidenced by comparing the performance of our deep tangency portfolio to standard machine learning portfolios, whose weights are a general function of characteristics. Our framework integrates a dense array of multi-market signals to endogenously capture risk premiums tied to liquidity frictions, shifts in credit quality, and term structure dynamics —

key economic drivers that are often fragmented in the decentralized corporate bond market.

Third, our analysis of the deep tangency portfolio highlights the importance of diverse firm characteristics in portfolio construction. We find: (i) the deep characteristic incorporates a broad set of firm characteristics, not just a narrow subset, providing little support for the sparsity often assumed in asset pricing and portfolio construction; and (ii) a broad multi-market characteristics set is essential, as the out-of-sample Sharpe ratio of the deep tangency portfolio falls to 1.01 without option variables and becomes even smaller with only bond or equity characteristics used. These results suggest that spanning the corporate bond efficient frontier requires a wide range of characteristics, contrasting with studies questioning the relevance of equity characteristics to bond returns (e.g., [Chordia, Goyal, Nozawa, Subrahmanyam, and Tong, 2017](#); [Bali, Goyal, Huang, Jiang, and Wen, 2021](#)), while aligning with evidence of their predictive power (e.g., [Dickerson et al., 2023](#)) and the bond-equity market integration (e.g., [Schaefer and Strebulaev, 2008](#); [Kelly, Palhares, and Pruitt, 2023](#)).

Our paper contributes to the literature on robust portfolio construction by avoiding the direct estimation of expected returns and covariance matrices. Early studies, such as [Brandt \(1999\)](#) and [Brandt et al. \(2009\)](#), introduce nonparametric and parametric approaches to estimate portfolio weights, thereby bypassing the need to estimate the return distribution. However, these methods struggle with a large number of characteristics or assets and cannot capture nonlinear relationships, reflecting key strengths of our deep parametric portfolio framework.

Our approach also aligns with recent advancements in machine learning (ML) for asset pricing and portfolio construction. Studies highlight the role of nonlinear, dense characteristic structures in modeling cross-sectional returns (see, among others, [Freyberger, Neuhierl, and Weber, 2020](#); [Gu, Kelly, and Xiu, 2020](#); [Giannone, Lenza, and](#)

Primiceri, 2021).³ More recent works integrate economic constraints into ML methods with a Sharpe ratio objective similar to ours, such as Gujarro-Ordóñez et al. (2025) with statistical arbitrage via convolutional transformers, Jensen, Kelly, Malamud, and Pedersen (2026) with trading-cost-aware optimization, Liu, Zhou, and Zhu (2024) with a genetic programming approach, and deep learning innovations in Chen, Pelger, and Zhu (2024), Cong, Tang, Wang, and Zhang (2022), and Simon, Weibels, and Zimmermann (2026), alongside the decision tree approach by Bryzgalova et al. (2025) who employ a nonparametric method to construct max-Sharpe ratio portfolios using long-only baseline portfolios and also compare the effect of different types of objective functions (prediction vs. Sharpe ratio objectives).

While most of these works estimate portfolio weights as a general function of characteristics, ours — positioned as an investment “enhancement device” — differs in that we explicitly introduce benchmark factors as our economic priors and impose a nonlinear long-short structure on the deep factor, derived from high-dimensional characteristics. Empirically, we show that these structural designs have an essential impact on portfolio performance.

By focusing on the corporate bond market, where cross-sectional pricing is less understood, our paper makes a significant contribution to asset pricing and investment in this market. Unlike recent machine learning studies focusing on forecasting bond returns (e.g., Bali et al., 2021; Feng, He, Wang, and Wu, 2025; Avramov, Jostova, and Philipov, 2025), we develop a flexible deep learning framework positioned as an enhancement device and show that it outperforms standard machine learning approaches, thereby shedding light on how information shapes the mean-variance frontier in the corporate bond market.

The remainder of the paper explores insights into asset pricing and practical investment strategies. Section 2 outlines our parametric portfolio approach, which integrates

³See Nagel (2021) and Kelly and Xiu (2023) for recent surveys.

high-dimensional data into the portfolio construction process. Section 3 presents corporate bond return data and firm characteristics used in the analysis. Section 4 assesses the empirical performance relative to existing methods, with an emphasis on enhancements in risk-adjusted returns. Section 5 concludes with implications for asset pricing and investment practice. Additional results are provided in the Online Appendix.

2 Methodology

2.1 Maximum Sharpe Ratio Portfolios

We anchor our methodology in the mean-variance efficiency framework, adapting the classic model of Markowitz (1952) to high-dimensional corporate bond data. Consider N assets in month t with excess returns $r_t = [r_{1,t}, \dots, r_{N,t}]'$. The MVE portfolio, optimal for risk-averse investors seeking maximum risk-adjusted returns, is given by:

$$R_t^{opt} = w_{t-1}' r_t, \quad (1)$$

with the portfolio weights w_{t-1} being:

$$w_{t-1} = \Sigma_{t-1}^{-1} \mu_{t-1}, \quad (2)$$

where $\mu_{t-1} = E_{t-1}[r_t]$ is the conditional expected excess return, and $\Sigma_{t-1} = Cov_{t-1}(r_t)$ is the conditional covariance matrix. The maximum conditional squared Sharpe ratio of the optimal portfolio is defined as:

$$SR_{t-1}^2(R_t^{opt}) = \mu_{t-1}' \Sigma_{t-1}^{-1} \mu_{t-1}. \quad (3)$$

While the mean-variance framework is a cornerstone of portfolio construction, it poses significant practical challenges for investors. Estimating expected returns (μ_{t-1}) is often highly imprecise, and in a large asset universe, computing the covariance matrix (Σ_{t-1}) can be computationally intensive and prone to error. Consequently, both challenges may lead to unreliable portfolio weights, resulting in suboptimal out-of-sample performance for mean-variance efficient portfolios (see, e.g., Kan and Zhou,

2007; DeMiguel, Garlappi, and Uppal, 2009). Instead of relying on noisy estimates of expected returns (μ_{t-1}) and covariance matrices (Σ_{t-1}), we consider directly estimating the portfolio weights (see, e.g., Brandt, Santa-Clara, and Valkanov, 2009).

A conventional approach models the portfolio weights w_{t-1} as a linear combination of the market-capitalization weights and a characteristic-managed component that is a linear function of J characteristics (z_{t-1}), where $J \ll N$:

$$w_{t-1} = \tilde{w}_{t-1} + z_{t-1}\kappa, \quad (4)$$

where $\tilde{w}_{t-1}$ is market-capitalization weights, z_{t-1} is the cross-sectionally standardized characteristics, and κ is a $J \times 1$ coefficient vector. Let $R_{m,t} = \tilde{w}_{t-1}'r_t$, representing the market portfolio, and $f_t = z_{t-1}'r_t$, representing J characteristic-managed portfolios. The optimal portfolio in Equation (1) then becomes:

$$R_t^{opt} = R_{m,t} + \kappa'f_t = \delta F_t, \quad (5)$$

where $\delta = [1, \kappa']$ and $F_t = [R_{m,t}, f_t']'$. Notably, the maximum conditional squared Sharpe ratio in Equation (3) now equals that of the factor set F_t , i.e., $SR_{t-1}^2(R_t^{opt}) = \mu_{F,t-1}'\Sigma_{F,t-1}^{-1}\mu_{F,t-1}$, allowing investors to evaluate the portfolio performance using commonly used benchmark factors.

Two critical challenges arise with the conventional approach in Equation (4): (i) by only focusing on a narrow set of characteristics, it misses many valuable return signals; and (ii) it fails to account for complex, nonlinear relations among characteristics that may boost the portfolio performance. In this paper, we develop a deep learning framework designed to address these two issues. We directly construct the tangency portfolio weights as a complex function of a large set of K characteristics ($z_{i,t-1}$), where $K \gg J$, thereby covering thousands of individual assets. We define the portfolio weights, $w_{i,t-1}$, as follows:

$$w_{i,t-1} = \tilde{w}_{i,t-1} + \theta w_d(z_{i,t-1}; \Phi), \quad i = 1, \dots, N, \quad (6)$$

where, as before, $\tilde{w}_{i,t-1}$ is the weight of asset i in the market portfolio, $w_d(\cdot)$ is a function of $z_{i,t-1}$ that can account for any potential nonlinear relations among a large number of characteristics of asset i and generates the weights for constructing a zero-cost long-short portfolio (to be explained in detail in the next subsection), Φ is the required parameters, and θ is a scalar controlling the relative weight in the tangency portfolio.

Then, the optimal portfolio return in Equation (1) can be represented by:

$$R_t^{opt} = \sum_{i=1}^N \tilde{w}_{i,t-1} r_{i,t} + \theta \sum_{i=1}^N w_d(z_{i,t-1}; \Phi) r_{i,t} = R_{m,t} + \theta R_{d,t}, \quad (7)$$

where $R_{m,t}$, again, is the market portfolio return, and given that $w_d(\cdot)$ cross-sectionally sums to zero, $R_{d,t}$ is, in fact, the return on a long-short portfolio based on non-linear combinations of characteristics. $R_{d,t}$ complements the market factor, which represents a benchmark strategy, by employing information contained in a large number of characteristics. When the function $w_d(\cdot)$ takes a linear form, and the number of characteristics is set to be small, our parameterization becomes the conventional approach as in Equation (4).

When we have *a priori* knowledge that a particular set of observable factors helps span the efficient frontier, we can introduce these factors by inserting them into Equation (6) and construct the portfolio weights, $w_{i,t-1}$, as follows:

$$w_{i,t-1} = \tilde{w}_{i,t-1} + \tilde{w}_{i,t-1}^p \theta_p + \theta_d w_d(z_{i,t-1}; \Phi), \quad i = 1, \dots, N, \quad (8)$$

where $\tilde{w}_{i,t-1}^p$ is a $1 \times P$ vector of weights on individual assets for constructing the P observable factors, and θ_p is a $P \times 1$ vector of coefficients. The optimal portfolio return is then given by:

$$R_t^{opt} = R_{m,t} + \theta_p' R_{p,t} + \theta_d R_{d,t}, \quad (9)$$

where $R_{p,t}$ is a $P \times 1$ vector of returns on P observable factors at time t . Now, the θ in Equation (7) becomes $\theta = [\theta_p', \theta_d]'$.

The primary objective of our framework is to construct a tangency portfolio with the maximum Sharpe ratio, consistent with the mean-variance efficiency objective. To this end, we define the function $w_d(\cdot)$ and estimate model parameters (θ, Φ) by maximizing the average conditional squared Sharpe ratio of the optimal portfolio R_t^{opt} :

$$\max_{\theta, \Phi} \frac{1}{T} \sum_{t=1}^T S R_{t-1}^2(R_t^{opt}). \quad (10)$$

Guided by the principle of diversification, this optimization yields clear investment insights, enabling the long-short deep factor ($R_{d,t}$) to exhibit a low or even negative correlation with benchmark factors (e.g., the market factor), thereby delivering a hedge portfolio.

Our approach provides a practical tool to enhance the risk-adjusted returns of the optimal portfolio without relying on noisy estimates of expected returns and covariance matrices. Previous empirical work shows that even well-established factors fail to fully capture both cross-sectional and time-series variation in asset returns. We employ high-dimensional features across thousands of assets, capturing intricate nonlinear signals that these benchmark factors often overlook. Nevertheless, the benchmark factors often contain valuable pricing information. Therefore, our approach, as seen in Equations (7) and (9), can be viewed as an investment “enhancement device” that constructs the tangency portfolio based on these benchmark factors, which serve as our “prior” information.

2.2 Deep Factor and Deep Tangency Portfolio

We now detail the deep learning framework for constructing the portfolio weights $w_d(\cdot)$ and the associated deep long-short portfolio $R_{d,t}$. While established characteristic-managed factors predominantly rely on sparse linear specifications, they may neglect important nonlinear signals that could prove essential for effective portfolio construction. Our approach uses a large number of characteristics and their nonlinear interac-

tions to extract characteristic-based components that improve benchmark portfolios. We construct the long-short deep factor, $R_{d,t}$, using a neural network optimized for a Sharpe ratio objective, positioning it as a structural complement to existing benchmark factors. This target specifically focuses on maximizing the Sharpe ratio, while retaining the conventional sorting scheme in our deep learning framework.

Deep Characteristic. We follow the standard modeling approach of a neural network for dimensional reduction of a large number of characteristics (see, e.g., [Gu, Kelly, and Xiu, 2020](#); [Feng, He, Polson, and Xu, 2024](#)). To formalize the model, we define a training observation at time t as a set containing:

- $\{r_{i,t}\}_{i=1}^N$, excess returns of N individual assets;
- $\{z_{k,i,t-1} : 1 \leq k \leq K\}_{i=1}^N$, K characteristics of N assets observed at time $t - 1$;
- $\{R_{b,t}\}_{b=1}^{P+1}$, a $(P + 1) \times 1$ vector of excess returns on the market factor and P observable factors.

We design an L -layer neural network that transforms K characteristics into one deep characteristic that is relatively interpretable. At each time t and for each asset i ($i = 1, \dots, N$), our deep parametric portfolio approach works as follows,

$$Z_{i,t-1}^{(0)} = [z_{1,i,t-1}, \dots, z_{K,i,t-1}]', \quad (11)$$

$$Z_{i,t-1}^{(l)} = G(A^{(l)} Z_{i,t-1}^{(l-1)} + b^{(l)}), \quad (12)$$

for $l = 1, \dots, L$, where $Z_{i,t-1}^{(l)}$ is the i -th column of the $K_l \times N$ matrix of $Z_{t-1}^{(l)}$, for $1 \leq K_l \leq K$, and $G(\cdot)$ is an element-wise activation function, which is chosen to be the $\tanh$ function in the paper, $G(x) = (e^x - e^{-x}) / (e^x + e^{-x})$. $A^{(l)}$ and $b^{(l)}$ are deep learning weights and biases, respectively, and need to be trained during the algorithm.

The algorithm performs the transformation and dimension reduction for each asset independently, without interactions among different assets, using a univariate activa-

tion function. In the end, we have a $1 \times N$ matrix of deep characteristics, $Z_{t-1}^{(L)}$. The parameters to be trained in this part are deep learning weights A and biases b , namely,

$$\left\{ (A^{(l)}, b^{(l)}) : A^{(l)} \in \mathbb{R}^{K_l \times K_{l-1}}, b^{(l)} \in \mathbb{R}^{K_l} \right\}_{l=1}^L. \quad (13)$$

Deep Factors. The deep characteristic, $Z_{t-1}^{(L)}$, is then used to form weights of a deep portfolio (factor) as follows:

$$w_d(z_{t-1}) \equiv W_{t-1} = h(Z_{t-1}^{(L)}), \quad (14)$$

where the function, $h(\cdot)$, needs to be differentiable.

Following the literature, our first choice of the function $h(\cdot)$ is simply a linear function, resulting in a deep characteristic-managed portfolio with a scaling parameter a ,

$$W_{t-1} = \frac{a}{N} Z_{t-1}^{(L)}. \quad (15)$$

Moreover, to mimic the commonly used portfolio-sort approach (i.e., a nondifferentiable step function), following [Feng et al. \(2024\)](#), we adopt the *softmax* function and compute the portfolio weights as follows. For $x = Z_{t-1}^{(L)}$, the function $h(\cdot)$ takes the form of,

$$h(x) = \begin{bmatrix} \text{softmax}(x_1^+) \\ \text{softmax}(x_2^+) \\ \vdots \\ \text{softmax}(x_N^+) \end{bmatrix} - \begin{bmatrix} \text{softmax}(x_1^-) \\ \text{softmax}(x_2^-) \\ \vdots \\ \text{softmax}(x_N^-) \end{bmatrix}, \quad (16)$$

where $x_i^+ := -a_1 e^{-a_2 x_i}$ and $x_i^- := -a_1 e^{a_2 x_i}$ for each element i , and a_1 and a_2 are two tuning parameters. The nonlinear *softmax* function is an increasing function,

$$\text{softmax}(x_i) = \frac{e^{x_i}}{\sum_{j=1}^N e^{x_j}}, \quad (17)$$

and $\sum_{i=1}^N \text{softmax}(x_i) = 1$. On the right-hand side of Equation (16), the first term represents the long position weights of assets, and the second term is for the symmetric

short position. In implementation, we tune the two hyperparameters, a_1 and a_2 , and choose $a_1 = 50$ and $a_2 = 8$. For these two values, we note that approximately 50 % to 70% of assets are in the middle rank and have negligible weights, similar to the traditional sorting procedure (see Figure A2 in the Online Appendix). Furthermore, we normalize the portfolio weights so that the sum of the long-leg weights equals 1 and the sum of the short-leg weights equals -1. As discussed in Feng et al. (2024), this nonlinear ranking scheme depends on the cross-sectional rank information and the distributional properties of characteristics. Such a construction of the deep factor avoids extreme positions in both long and short legs (Avramov et al., 2023).

In what follows, we refer to Equation (15) as the linear ranking and to Equation (16) as the softmax ranking. The deep factor portfolio weights, W_{t-1} , in Equation (15) or (16), sum to zero by construction. Our deep factor, $R_{d,t}$, can then be computed as:

$$R_{d,t} = W_{t-1}r_t. \quad (18)$$

Loss Function. The deep factor in Equation (18) can be combined with the market or other benchmark factors to form the deep tangency portfolio as in Equation (7) or (9). Given that the model parameters are estimated to be time-invariant, and assuming that the high-dimensional characteristics $z_{i,t-1}$ fully capture the conditional expected returns and covariance structure relevant for the optimal portfolio, the objective function in Equation (10) simplifies from maximizing the average conditional squared Sharpe ratio to maximizing the unconditional squared Sharpe ratio of the optimal portfolio R_t^{opt} on $\tilde{F}_t = [R'_{b,t}, R_{d,t}]'$,

$$SR^2(R_t^{opt}) \equiv SR^2(\tilde{F}_t) = \mathbf{E}(\tilde{F}_t)' \mathbf{Cov}(\tilde{F}_t)^{-1} \mathbf{E}(\tilde{F}_t). \quad (19)$$

A multi-layer neural network usually has a large number of parameters. To avoid overfitting and improve the model's OOS performance, we augment the objective function with mini-batch stochastic gradient descent and dropout regularization, which, in general, performs better than l_1/l_2 regularization in neural networks (Chen et al.,

2024).⁴ The neural network is trained by minimizing the following loss function:

$$\mathcal{L} = \mathbb{E}_{B \sim \mathcal{D}_B, D \sim \text{Dropout}(p)} \left\{ \exp \left[-SR^2 \left(R_t^{opt}(B, D) \right) \right] \right\}, \quad (20)$$

where $B \sim \mathcal{D}_B$ denotes a randomly sampled mini-batch from the training data, and $D \sim \text{Dropout}(p)$ represents a stochastic dropout mask with probability p . To ensure numerical stability and prevent overfitting in our high-dimensional setting, we employ mini-batch stochastic gradient descent in conjunction with dropout regularization. These standard safeguards are critical for training a multi-layer architecture on the relatively noisy cross section of corporate bond returns. We emphasize that our deep parametric portfolio approach is not merely a return-prediction model. Instead, it functions as a long-short factor automatic generation model, where the spread between the long and short tails of about 30% of the observations is not easily overfitted by transforming the high-dimensional characteristics into a single univariate variable for sorting.

The hyperparameters, such as the batch size $\mathcal{D}_B$ and the dropout rate p , need to be tuned using training and validation data. Figure 1 presents a visualization of our deep learning architecture, summarizing the critical stages for constructing the deep factor and the deep tangency portfolio.

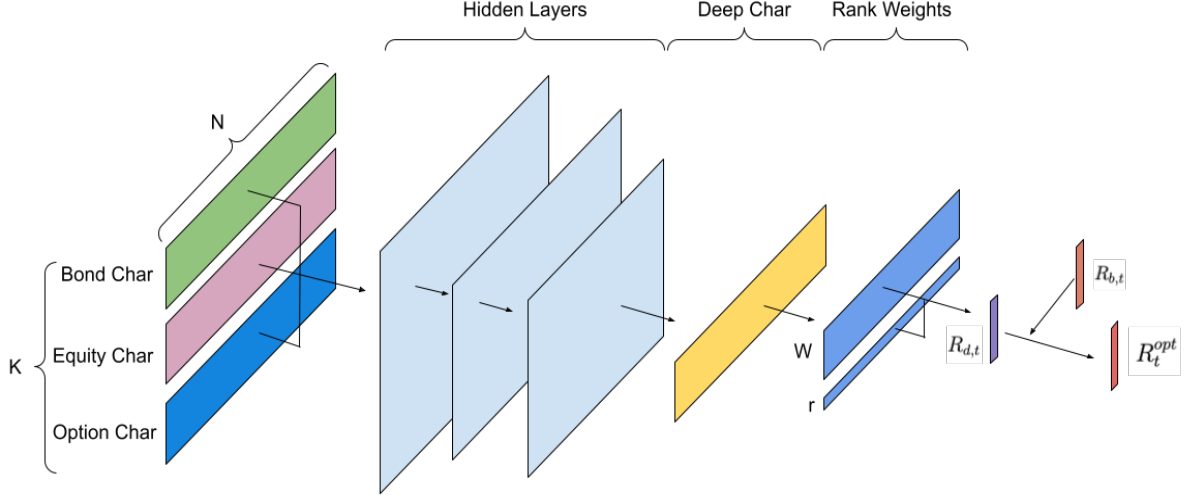
3 Data

We assess the framework’s empirical utility within the U.S. corporate bond market, as studies on the cross-sectional pricing of corporate bonds relative to the equity market remain limited. We first construct corporate bond returns from TRACE data in Section 3.1; Section 3.2 details the multi-market characteristics used in our deep parametric architecture; and we present the benchmark factors in Section 3.3.

⁴For foundational discussions of advantages of dropout over classical norm-based regularization, see the textbook *Deep Learning* by Goodfellow, Bengio, and Courville (2016).

Figure 1: Deep Learning Network Architecture

This figure provides a visualization of the deep learning architecture. Different types of characteristics, $Z^{(0)}$ (e.g., equity, bond, and option characteristics), are transformed via the multi-layer neural network to deep characteristics, $Z^{(L)}$, based on which the deep portfolio (factor) weights, W , are formed. The deep tangency portfolio, R_t^{opt} , is constructed by combining the deep factor, $R_{d,t}$, and the benchmark factors, $R_{b,t}$.



3.1 Corporate Bond Returns and Summary Statistics

We construct corporate bond return data using two authoritative sources for accuracy, aligning with institutional bond markets: TRACE for intraday transaction data, and the Mergent Fixed Income Securities Database (FISD) for bond-level characteristics. TRACE represents the regulatory reporting system for U.S. corporate bond pricing, capturing nearly all OTC trades and mitigating index-based biases (e.g., ICE BofA), which aligns with recent advances in bond asset pricing (Kelly and Pruitt, 2022; Dickerson et al., 2023). Integrating TRACE with FISD provides key characteristics, including offering date, offering amount, maturity date, coupon type and rate, bond type and rating, interest payment frequency, issuer information, and others, which are crucial for filtering and return computation.

Adhering to the standard data-cleaning protocols of Dick-Nielsen (2009, 2014),⁵ we

⁵Due to different versions of data cleaning processes in recent literature, we draw on more detailed and reasonable paths according to some common sources, such as the WRDS Bond database (detailed explanations in Appendix C of Jostova, Nikolova, and Philipov, 2024), and the Open Source Bond Asset

remove duplicates and erroneous trades in the TRACE data. Additionally, we apply several filters to the data such that we remove: (i) bonds that are not listed or traded in the U.S. public market; (ii) bonds that are structured notes, mortgage-backed, asset-backed, agency-backed, U.S. Government, or equity-linked; (iii) convertible bonds whose option feature distorts the return calculation and makes it impossible to compare the returns of convertible and nonconvertible bonds; (iv) bonds with time to maturity of less than two years; (v) bonds that trade under \$5 or above \$1,000; (vi) intraday transaction records that have trading volume less than \$10,000; and (vii) all bond transactions that have more than a two-day settlement. We then calculate the daily bond price as the trading-volume-weighted average of intraday prices, as in [Bessembinder, Kahle, Maxwell, and Xu \(2008\)](#). For monthly returns, we follow standard practice to capture total returns, including accrued interest and coupons ([Dickerson et al., 2023](#)). For bond i in month t , raw returns are computed as:

$$\tilde{r}_{i,t} = \frac{P_{i,t} + AI_{i,t} + C_{i,t}}{P_{i,t-1} + AI_{i,t-1}} - 1, \quad (21)$$

where $P_{i,t}$ is the month-end price, $AI_{i,t}$ represents accrued interest (income between coupons), and $C_{i,t}$ is coupon payments. To ensure timing consistency, we define “month-end” as the last five trading days of t (using the final observation for multiple trades) and “month-start” as the first five days of t , prioritizing returns from “end-of- $t-1$ to end-of- t ”. This approach avoids overlapping holding periods. Excess returns are then calculated as $r_{i,t} = \tilde{r}_{i,t} - r_{f,t}$, where $r_{f,t}$ is the 1-month Treasury bill rate from CRSP. This adjustment accounts for the risk-free rate, providing a standardized measure of bond performance relative to a baseline return.

To mitigate the small size or illiquidity issue, a known source of spurious results (see, e.g., [Kelly and Pruitt, 2022](#)), we construct a cross-sectionally consistent sample of

Pricing code repository.

the 3,000 largest bonds by market value (i.e., outstanding shares multiplied by month-end price) on a monthly basis. This choice is motivated by liquidity considerations: the 3,000 largest bonds exhibit 50% narrower bid-ask spreads and 70% fewer zero-trading days compared to smaller bonds. We note that over the sample period, 10 months have fewer than 3,000 eligible bonds (though never fewer than 2,800). For these months, we impute missing excess returns and characteristics (see Section 3.2) as 0. The final sample spans the period from July 2004 to December 2020 (including the 2008 crisis and the 2020 COVID-19 shock) and comprises 14,796 unique bonds and 590,991 bond-month excess return observations.

The summary statistics in Table 1 exhibit patterns consistent with the established corporate bond pricing literature. The full sample has a mean monthly excess return of 0.45% (annualized 5.4%) and a standard deviation of 3.41%, indicating a moderate risk-return profile for corporate bonds. Average bond size: \$846 million; average credit rating: 8.70 (numerical score = BBB+).⁶ Investment-grade (IG) bonds (75.7% of observations) have lower returns (0.42% vs. 0.56%) and volatility (2.61% vs. 5.13%) than non-investment-grade (NIG) bonds, consistent with credit risk compensation (Bai, Bali, and Wen, 2019). Public bonds (77.3% of observations) have higher ratings (8.27 vs. 10.18) and lower returns (0.44% vs. 0.51%) than private bonds, reflecting the greater transparency/liquidity of public debt (Jostova et al., 2024). Panels B and C show that high-rated bonds (AAA–A) are mostly long-maturity (>10 years), confirming the representativeness of the sample in the U.S. corporate bond market.

3.2 Characteristics

We utilize three distinct domains of characteristics to capture predictive signals for corporate bond returns. The first type includes 41 bond characteristics, grouped into three major categories: basis characteristics (e.g., rating, duration, liquidity), return-

⁶Credit ratings are converted to numerical scores (1 = AAA, 2 = AA+, ..., 21 = C) using S&P/Moody's ratings (averaged if both exist). IG bonds = scores 1–10 (BBB- or higher); NIG bonds = scores ≥ 11 .

Table 1: Summary Statistics

The table reports summary statistics of the whole sample (ALL) and several subsamples constructed based on Rating (Investment Grade (IG) & Non-Investment Grade (NIG)), ownership (Public & Private), and/or Maturity. The data contain 590,991 monthly return observations of 14,796 unique corporate bonds from July 2004 to December 2020.

Panel A: Cross-Sectional Statistics					
	ALL	IG	NIG	Public	Private
Bond-month observations	590,991	447,603	143,388	456,873	134,118
Return mean (%)	0.45	0.42	0.56	0.44	0.51
Return std (%)	3.41	2.61	5.13	3.15	4.14
Rating mean	8.70	7.02	13.95	8.27	10.18
Duration mean	3.98	4.27	3.09	4.07	3.69
Age mean	4.16	4.24	3.93	4.13	4.26
Size mean (\$ million)	846	900	678	871	764

Panel B: Sample Distribution (%) by Maturity and Rating						
Maturity	AAA	AA	A	B	Junk	ALL
2	0.16	0.74	2.65	2.60	1.37	7.52
3	0.20	0.81	3.04	3.24	1.93	9.23
4	0.18	0.81	3.10	3.54	2.46	10.09
5	0.16	0.80	3.15	3.78	3.04	10.93
6	0.10	0.42	1.97	2.86	3.35	8.69
7	0.09	0.41	1.95	2.93	3.38	8.77
8	0.08	0.36	1.79	2.78	2.64	7.64
9	0.07	0.36	1.80	2.84	1.93	7.01
10	0.07	0.38	1.90	2.93	1.34	6.62
≥11	0.52	1.48	8.47	10.23	2.82	23.51
ALL	1.62	6.57	29.81	37.74	24.26	100.00

Panel C: Sample Distribution (%) by Ownership and Rating						
Ownership	AAA	AA	A	B	Junk	ALL
Private	0.11	1.23	4.68	8.33	8.35	22.69
Public	1.50	5.34	25.13	29.41	15.92	77.31
ALL	1.62	6.57	29.81	37.74	24.26	100.00

distribution characteristics (e.g., momentum, reversal, variance, skewness), and co-variance characteristics (e.g., market beta, TERM beta, DEF beta).

Furthermore, given that both bond and stock prices are contingent on firm fundamentals, we also consider equity characteristics that have been shown to help predict

equity returns. Recent studies have shown that bond and equity markets are largely integrated. [Choi and Kim \(2018\)](#) argue that market integration suggests different markets should share common factors. [Schaefer and Strebulaev \(2008\)](#) show that bond and equity returns are related through the capital structure hedge ratio. By approximating the hedge ratio with a Merton model for debt, they find that the sensitivity of debt returns to equity is close to that predicted by the Merton model. Building on [Schaefer and Strebulaev \(2008\)](#) and [Choi and Kim \(2018\)](#), [Kelly et al. \(2023\)](#) find that debt and equity markets are more integrated than previous estimates suggest, and that these markets are substantially more integrated in terms of their systematic risks than their idiosyncratic risks. Therefore, the second type includes 61 equity characteristics commonly used in the literature, which cover six major categories: momentum, value, investment, profitability, size, and intangibles.

In addition, the recent literature has found that several option-related variables have predictive power for corporate bond returns (see, e.g., [Cao et al. 2023](#); [Chung et al. 2019](#); [Huang et al. 2023](#)). We, therefore, construct a total of 30 option-related characteristics. Many of those option-related variables have been shown to have predictive power for equity returns (see, e.g., [Neuhierl, Tang, Varneskov, and Zhou, 2021](#)); here, we examine whether they also help forecast corporate bond returns.

Altogether, we have a large number of characteristics (132 in total). The bond, equity, and option characteristics are listed in Tables [A1](#), [A2](#), and [A3](#), respectively, in the Online Appendix. Before feeding those characteristics into our deep parametric portfolio approach, we cross-sectionally rank and standardize them each month so that they fall within the range $[-1, 1]$ and have cross-sectional averages of 0. Any missing values are imputed to be 0. One advantage of using the cross-sectional ranks of characteristics is that the impact of potential data errors and outliers in individual characteristics can be largely alleviated (see, e.g., [Kelly et al., 2019](#); [Kozak et al., 2020](#)).

3.3 Benchmark Factors and Alternative Portfolios

Benchmark Market Factors. Recent studies in the corporate bond market show that the market factor is the most important one and explains most of the characteristic-managed bond factors (see, e.g., [Dickerson et al., 2023](#)). Therefore, we take both the equal- and value-weighted corporate bond market portfolios as our benchmarks in the main empirical analysis.

Alternative Portfolios. We consider alternative portfolios spanned by both observable and latent factors. The observable factors include the Fama-French five factors: three bond factors and two equity factors, as well as two types of latent factors: the IPCA factors and the RP-PCA factors.

(i) The Fama-French five factors (FF5). We combine the Fama-French three-factor model, which includes the value-weighted bond market factor, the term factor, and the default factor ([Fama and French, 1993](#)), with two equity factors: the size factor (SMB) and the value factor (HML) ([Fama and French, 1996](#)). The term factor is constructed as the difference between the long-term government bond returns and the one-month Treasury bill rate, and the default factor is the difference between the long-term corporate bond returns and the long-term government bond returns. The equity size and value factors are obtained from Ken French’s online data library.

(ii) IPCA Factors. A recent paper by [Kelly, Palhares, and Pruitt \(2023\)](#) shows that a five-factor model based on the IPCA of [Kelly, Pruitt, and Su \(2019\)](#) outperforms commonly used observable factor models in pricing corporate bonds. We follow their IPCA approach and construct five corporate bond factors using our TRACE data and all three types of characteristics.

(iii) RP-PCA Factors. Following the procedure of [Lettau and Pelger \(2020\)](#), we construct five RP-PCA factors using 60 tercile portfolios based on 20 characteristics, which

turn out to be the most important variables contributing to the deep characteristic.⁷ See Figure 4 in Section 4.4 for these 20 characteristics.

4 Empirical Findings

In our empirical implementation, we split the sample into two parts: the subsample from July 2004 to June 2014 for model training and validation, and the subsample from July 2014 to December 2020 for OOS testing. This split is based on the assumption that the in-sample period is sufficiently long to include a range of market conditions, including the 2008 global financial crisis. We employ a two-fold deterministic cross-validation scheme to determine the optimal hyperparameters.⁸ Note that the number of neural network layers and the two parameters in the softmax ranking scheme (i.e., a_1 and a_2) are also tuning parameters. See Section II in the Online Appendix for implementation details. Section III in the Online Appendix presents results from alternative split and validation schemes for robustness checks.⁹

This section presents our empirical results, focusing on the deep factor’s ability to enhance the benchmark investment and comparing the performance of our deep tangency portfolio with that of alternative portfolios.

4.1 Deep Corporate Bond Factors

Table 2 presents summary statistics of the OOS deep factors. We use both the equal- and value-weighted corporate bond market factors as our benchmarks when

⁷Given that our sample size is relatively small (120 months in sample and 78 months out of sample), the number of portfolios used for constructing RP-PCA factors cannot be too large.

⁸To be more specific, the two-fold deterministic design divides the sample from July 2004 to June 2014 into two equal-length consecutive fold samples. We train our neural network on one fold and validate on the other, then repeat the process by swapping the folds. We then average the corresponding validation losses to identify the optimal hyperparameters. Finally, we refit the model on the entire sample (July 2004 to June 2014) using the selected hyperparameters.

⁹Specifically, we consider two alternative splits. The first is to use the sample from January 2011 to December 2020 for in-sample training and validation, and the sample from July 2004 to December 2010 for OOS testing, with the same two-fold validation scheme (Section III.1). The other is a random split scheme (Section III.2). Following Fama and French (2018), we randomly split the data into three groups sequentially over time. That is, for each quarter, the three-month data from that quarter are randomly assigned to three groups, and this process is repeated over time until the end of the sample. We use data from any two groups for training and validation, and data from the other group for the OOS test.

constructing deep factors. In implementation, we restrict the weights of the long and short legs to 1 and -1, respectively. All OOS results are based on the in-sample parameter estimates. The optimal number of layers is determined by training and validating on in-sample data. The 2-layer neural network is the preferred choice in most cases, a performance gain that persists across both the linear (Equation 15) and softmax (Equation 16) ranking specifications. However, for comparison, Panels A and B present deep factors constructed from the 1-, 2-, and 3-layer neural networks for the OOS period.

Both validation and out-of-sample tests confirm that the 2-layer neural network achieves the best performance across most settings. Specifically, under the softmax ranking scheme, the 2-layer deep factor produces the highest Sharpe ratios, reaching 1.49 when using the value-weighted market factor as the benchmark and 1.00 when using the equal-weighted market factor. The only exception arises in the linear ranking model with the equal-weighted market factor benchmark, where the 3-layer neural network slightly outperforms the 2-layer one. Nevertheless, the 2-layer model still delivers the highest Sharpe ratio of 1.12 when the value-weighted market factor is taken as a benchmark. We also find that the softmax ranking-based deep factor outperforms the linear ranking-based one on both benchmarks, suggesting that imposing this long-short nonlinear structure via the softmax ranking is important in our framework. Therefore, in what follows, we implement our empirical analysis primarily using the softmax ranking-based deep learning model.

The table also presents the mean with its Newey-West t -statistics (in parentheses), minimum, maximum, and maximum drawdown of the deep factor. Taking the 2-layer softmax ranking-based deep factor as an example, we see that the deep factor returns vary from -1.54% (min) to 2.37% (max), with the mean return of 0.16% ($t = 2.65$) when taking the equal-weighted market factor as the benchmark, and vary from -1.12% (min) to 1.52% (max), with the mean return of 0.19% ($t = 4.54$) when taking the value-weighted market factor as the benchmark. The maximum drawdowns are

Table 2: Deep Corporate Bond Factors

This table presents the out-of-sample descriptive statistics of the deep factor returns, including mean and its Newey-West t -statistics (in parentheses), the annualized Sharpe ratio, minimum, maximum, and maximum drawdown. Both equal- and value-weighted market factors are considered as benchmarks, along with linear and softmax ranking schemes. R^i denotes the deep factor return from a neural network with i layers. The sample from July 2004 to June 2014 is used for model training and validation, while the sample from July 2014 to December 2020 is for OOS testing.

	Equal-Weighted Market Factor					Value-Weighted Market Factor				
	Mean	SR	Min	Max	Max DD	Mean	SR	Min	Max	Max DD
Panel A: Deep Factors: Linear Ranking										
R_l^1	0.04 (0.85)	0.38	-1.05	0.94	3.21	0.10 (1.67)	0.76	-1.20	1.33	2.55
R_l^2	0.04 (0.70)	0.34	-1.77	1.26	3.54	0.14 (2.38)	1.12	-1.28	1.22	2.06
R_l^3	0.05 (1.20)	0.55	-0.91	1.00	2.28	0.11 (1.99)	0.95	-0.78	1.42	2.11
Panel B: Deep Factors: Softmax Ranking										
R_s^1	0.12 (1.20)	0.55	-1.83	2.25	5.22	0.17 (2.65)	1.21	-1.00	1.37	2.26
R_s^2	0.16 (2.65)	1.00	-1.54	2.37	2.48	0.19 (4.54)	1.49	-1.12	1.52	1.71
R_s^3	0.14 (1.52)	0.71	-1.63	2.16	4.31	0.05 (0.21)	0.09	-9.44	5.90	17.50

approximately 2.48% and 1.71% for both benchmarks, which are relatively small compared to the standard bond market factors, further indicating superior risk-adjusted performance.

4.2 Deep Tangency Portfolios

We next evaluate the performance of the deep tangency portfolios. Panel A of Table 3 presents annualized Sharpe ratios of our deep tangency portfolios for the OOS period. Note that all portfolios' weights are determined by the in-sample estimates, and the optimal number of neural network layers tuned by training and validation equals 2. When using the equal-weighted market factor as the benchmark, the deep tangency portfolio achieves the highest annualized OOS Sharpe ratio of 2.13 in the 2-layer softmax ranking-based deep learning model, with a maximum drawdown of 0.89% (in square brackets). When we use the value-weighted market factor as the benchmark, the deep tangency portfolio delivers a similar OOS Sharpe ratio of 2.21. The Sharpe ratios of the deep tangency portfolios are significantly higher than those

Table 3: Deep Tangency Portfolios and Alternative Portfolios

This table presents the out-of-sample annualized Sharpe ratios of various portfolios for the out-of-sample period. Panel A presents the deep tangency portfolio, which uses either the equal-weighted or value-weighted market factor as its benchmark and employs the softmax ranking. L_i represents a neural network with i layers. Panel B presents the Sharpe ratios of alternative portfolios spanned by the Fama-French five factors, IPCA five factors, and RP-PCA five factors, respectively (row “SR”), and of portfolios spanned by these different types of factors together with the deep factor (row “SR (+DF)”) constructed using the equal-weighted market factor as the benchmark and the softmax ranking. In square brackets, the maximum drawdowns are reported in percentage. We follow [Barillas and Shanken \(2017\)](#) to test the significance of the Sharpe ratio improvement. In Panel A, we test the Sharpe ratio improvements of each deep tangency portfolio relative to the benchmark market factor, and in Panel B, we test the improvement in SR (+DF) over SR. ***, **, and * denote the levels of significance of 1%, 5%, and 10%, respectively. The sample from July 2004 to June 2014 is used for model training and validation, and the sample from July 2014 to December 2020 is for out-of-sample testing.

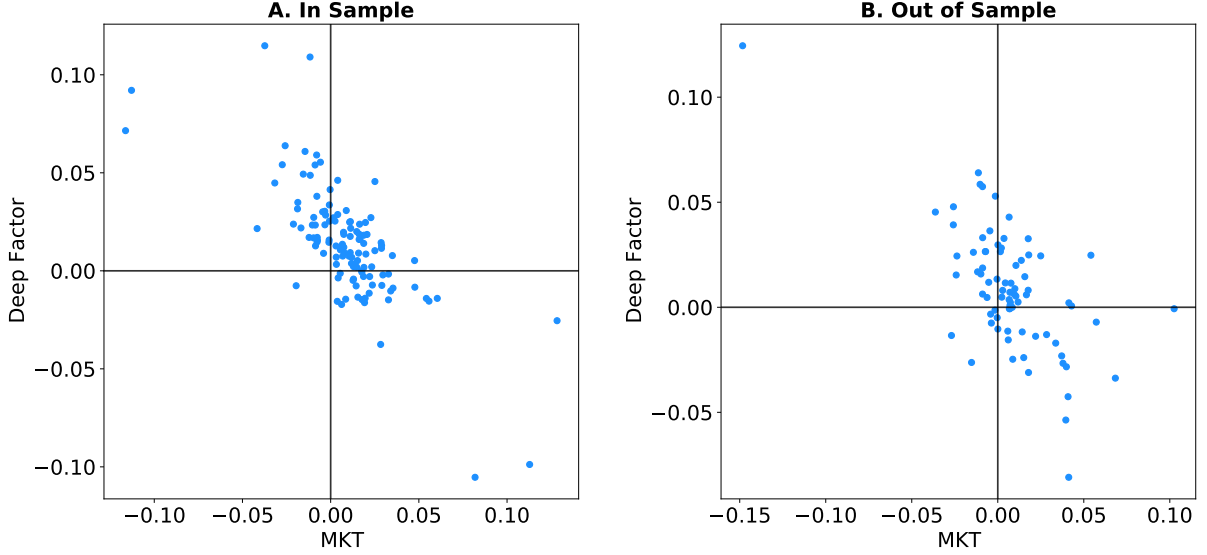
Panel A: Deep Tangency Portfolios					Panel B: Alternative Portfolios			
	MKT	L_1	L_2	L_3		FF5	IPCA5	RP-PCA5
Equal-Weighted	0.85	0.80	2.13*** [0.89]	0.95***	SR	0.88 [2.98]	1.48 [3.07]	0.93 [4.35]
Value-Weighted	0.90	1.43***	2.21*** [0.88]	1.08***	SR (+DF)	1.89*** [0.93]	1.91*** [1.09]	2.04*** [1.04]

of the corresponding market factors, and the improvements in Sharpe ratio are highly statistically significant, as indicated by the test of [Barillas and Shanken \(2017\)](#).

The high Sharpe ratio of the deep tangency portfolio is attributable to the negative correlation between the market factor and the deep factor in our trained deep learning model. This negative correlation enhances diversification, providing investors with a robust hedge against market downturns and reducing overall portfolio volatility. Panel A of Figure 2 presents the scatter plot between the equal-weighted market factor and the 2-layer softmax ranking-based deep factor for the in-sample period, showing a strong negative correlation. Panel B of Figure 2 presents the same scatter plot for the OOS period, confirming that the deep factor remains negatively correlated with the market factor. The scarcity of observations in the lower-left quadrant of Figure 2 confirms that the deep and market factors rarely experience joint drawdowns, empirically validating the model’s diversification objective.

Figure 2: Correlations between the Market and Deep Factors

Panels A and B present scatter plots between the bond market factor and the deep factor for the in-sample and out-of-sample periods. The deep factor is constructed using the equal-weighted market factor as the benchmark and the softmax ranking. The factor's in-sample annualized volatility is normalized to 10%, and its OOS returns are adjusted accordingly. The in-sample period ranges from July 2004 to June 2014, and the out-of-sample period from July 2014 to December 2020.

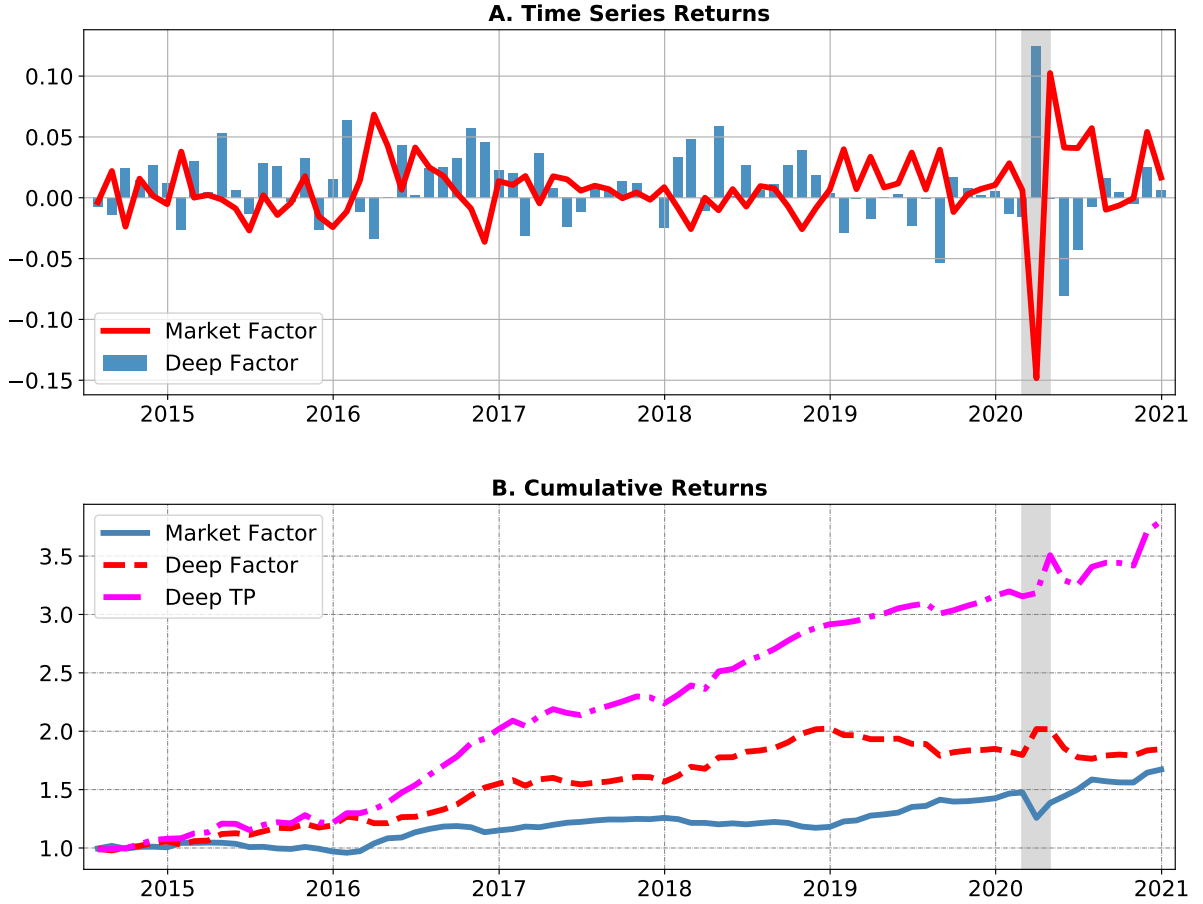


To further verify this point, Panel A of Figure 3 presents the time series of excess returns on the equal-weighted market factor and the deep factor over the OOS period. We see that the deep factor remains positive during periods of market downturns. More importantly, during the COVID-19 pandemic, the market factor declined sharply, whereas our deep factor generated a high positive return. Empirically, the deep factor serves as an effective market hedging portfolio. Panel B of Figure 3 displays the OOS cumulative returns of the equal-weighted market factor, the deep factor, and the deep tangency portfolio, respectively. Both the deep factor and the deep tangency portfolio outperform the market factor. We observe that during the COVID-19 pandemic, the cumulative returns of the deep factor and the deep tangency portfolio continue to increase, whereas the market factor suffers significant losses, consistent with the findings in Panel A.¹⁰

¹⁰The out-of-sample period ends in December 2020 and therefore does not cover the subsequent post-2021 rising-rate environment. Our evidence should thus be interpreted as demonstrating strong out-of-sample performance over July 2014–December 2020, including the COVID-19 stress episode, while leaving performance in a sustained rising-rate cycle to future research.

Figure 3: Time Series and Cumulative Returns

Panel A presents the time series returns of the market factor and the deep factor for the out-of-sample period, and Panel B presents the cumulative returns of the market factor, the deep factor, and the deep tangency portfolio for the out-of-sample period. All portfolios are normalized to have 10% annualized volatility. The deep factor and the deep tangency portfolio are constructed using the equal-weighted market factor as the benchmark and the softmax ranking. The out-of-sample period ranges from July 2014 to December 2020.



Panel B of Table 3 shows that the optimal portfolios constructed from commonly used observable and latent factors perform worse than our deep tangency portfolio for the same OOS period. As before, the portfolio weights are determined by the in-sample estimates. The optimal portfolio constructed from the Fama-French five factors has a Sharpe ratio of 0.88, which is significantly smaller than that of the deep tangency portfolio, and its maximum drawdown becomes much higher. Note that the Fama-French five-factor model includes two equity factors (SMB and HML). Then what happens when we combine the Fama-French factors and our deep factor? We observe that the annualized Sharpe ratio of the optimal portfolio, spanned by the

Fama-French five factors and our deep factor (with the equal-weighted market factor serving as the benchmark), increases dramatically (from 0.88 to 1.89), while the maximum drawdown drops to 0.93%.

A recent paper by [Kelly et al. \(2023\)](#) shows that a five-factor model based on the instrumental principal component analysis (IPCA, [Kelly et al., 2019](#)) outperforms commonly used observable factor models in pricing corporate bonds. They find that a tangency portfolio constructed from their five IPCA factors using ICE corporate bond return data can earn an annualized OOS Sharpe ratio of up to 6.23. We note that in another paper, [Kelly and Pruitt \(2022\)](#) show that the core analysis of [Kelly et al. \(2023\)](#) is robust to using the TRACE data. We follow their IPCA approach and construct five corporate bond factors using our TRACE data and all three types of characteristics. To be consistent with our primary empirical analysis, we use the same in-sample and OOS periods split as before and extract the OOS IPCA factors using in-sample model parameter estimates and OOS characteristics.

Even though both [Kelly et al. \(2023\)](#) and [Kelly and Pruitt \(2022\)](#) find that an optimal portfolio constructed using the five IPCA factors can earn an OOS Sharpe ratio of larger than 6 when using both ICE and TRACE corporate bond data, we find that such a portfolio can only earn an OOS Sharpe ratio of 1.48, which is much smaller than that of our deep tangency portfolio. There are two possible reasons for the weaker OOS Sharpe ratio we observe. First, the sample size in our paper is much larger than that in [Kelly and Pruitt \(2022\)](#): the total number of bond-month observations in our paper is 590,991, whereas it is only 144,933 in [Kelly and Pruitt \(2022\)](#). Second, both [Kelly et al. \(2023\)](#) and [Kelly and Pruitt \(2022\)](#) adopt an expanding window procedure to construct the OOS IPCA factors, whereas we extract OOS IPCA factors by fixing model parameters at the in-sample estimates to make it comparable with our methodology.¹¹

¹¹We also implement a recursive expanding-window approach similar to [Kelly et al. \(2023\)](#) and [Kelly and Pruitt \(2022\)](#) using our TRACE data and find an almost identical OOS Sharpe ratio of the IPCA optimal portfolio. The results show a mean of 0.39% (monthly) and a standard deviation of 0.81% (monthly), resulting in a Sharpe ratio of 1.67.

We further find that combining the deep factor with the IPCA five factors improves the OOS Sharpe ratio of the resulting optimal portfolio to 1.91, with a slightly higher maximum drawdown of 1.09%.

In addition, [Lettau and Pelger \(2020\)](#) propose a risk-premium principal component analysis (RP-PCA) model for estimating latent asset pricing factors. They demonstrate that RP-PCA outperforms PCA, particularly in identifying weak factors. We also examine how the five RP-PCA factors perform compared to our deep tangency portfolio. We construct the RP-PCA factors using 60 tercile portfolios based on the most important 20 characteristics, as discussed in Section 3.3. Again, we find that the optimal portfolio spanned by the five RP-PCA factors from the cross section of simple portfolio sorts has a much smaller OOS Sharpe ratio than our deep tangency portfolio, with a much higher maximum drawdown (4.35%).¹² When we combine the RP-PCA five factors with our deep factor, we find that the Sharpe ratio of the resulting optimal portfolio increases to 2.04, and its maximum drawdown decreases to 1.04%.

Moreover, to assess economic relevance, we evaluate the performance of our deep tangency portfolio with transaction costs included. As detailed in Section IV of the Online Appendix, Table A6 reports the Sharpe ratio of the deep tangency portfolio adjusted by representative transaction-cost assumptions of corporate bonds, proxied by the multiplications between different levels of the trading cost (15 or 20 basis points) and turnovers calculated from exponentially smoothed weights. The results show that our 2-layer deep tangency portfolio continues to perform best, confirming that our findings are robust to realistic trading frictions.

4.3 Factor-Spanning Regressions

Empirical analysis confirms that the deep factor endogenously serves as a market hedge, as its Sharpe-ratio maximization objective identifies cross-sectional signals that

¹²Because RP-PCA performance is contingent on the specific cross section of test portfolios, the results in Table 3 serve as a standardized comparison using the benchmark characteristic-sorted portfolios common in the literature.

are negatively correlated with benchmark drawdowns. Moreover, commonly used observable and latent factors do not add sufficient incremental information to improve the Sharpe ratio when combined with the deep factor. In this section, we further examine whether these commonly used observable and latent factors can explain excess returns on our deep factor and deep tangency portfolio by implementing factor-spanning regressions of the form,

$$R_t = \alpha + \beta' f_t + \epsilon_t, \quad (22)$$

where f_t is a set of observable or latent factors, and R_t represents returns on either the deep factor or the deep tangency portfolio. We use the equal-weighted bond market factor as the benchmark and employ a 2-layer neural network with a softmax ranking in our deep parametric portfolio approach.

Table 4 presents the factor-spanning regression results for the OOS period. Panel A employs the Fama-French five factors discussed in Section 3.3. We observe that these factors fail to fully explain the excess returns of both the deep factor (R_2^d) and the deep tangency portfolio (R_2^{opt}). The alpha estimates are 0.26% for the deep factor regression and 0.21% for the deep tangency portfolio regression. Both alpha estimates are highly statistically significant. The deep factor and the deep tangency portfolio load negatively on the size factor but positively on the value factor. However, the loadings on these two bond factors are not statistically significant. Both bond factors show negative loadings, although neither is statistically significant. The adjusted R^2 s are approximately 44.3% and 6.68% in the two spanning regressions.

Given that Kelly and Pruitt (2022) and Kelly et al. (2023) demonstrate that IPCA factors, estimated using all firm characteristics linearly, significantly outperform popular observable factors, we examine whether they can span our deep factor and deep tangency portfolio. The spanning regression results in Panel B show that the five IPCA factors cannot explain the excess returns on both the deep factor and the deep tan-

Table 4: Factor-Spanning Regressions

This table reports factor-spanning regression results for the out-of-sample period. Specifically, we regress the deep factor and the deep tangency portfolios on the Fama-French five factors, the IPCA five factors, and the RP-PCA five factors, respectively. The deep factor and the deep tangency portfolio are constructed using the equal-weighted market factor as the benchmark and the softmax ranking. The alpha estimates are expressed as percentages, and the Newey-West t -statistics are reported in brackets. The out-of-sample period ranges from July 2014 to December 2020.

Panel A: FF Five Factors							
	α	β_{MKT}	β_{SMB}	β_{HML}	β_{TRM}	β_{DEF}	Adj R^2 (%)
R_2^d	0.26 (5.99)	-0.09 (-1.28)	-0.03 (-1.25)	0.03 (1.41)	-0.07 (-1.48)	-0.12 (-1.55)	44.3
R_2^{opt}	0.21 (5.99)	0.11 (1.97)	-0.02 (-1.25)	0.02 (1.41)	-0.05 (-1.48)	-0.10 (-1.55)	6.68
Panel B: IPCA Five Factors							
	α	β_1	β_2	β_3	β_4	β_5	Adj R^2 (%)
R_2^d	0.21 (4.20)	-0.09 (-6.61)	0.09 (2.14)	-0.09 (-2.50)	0.05 (1.36)	0.08 (0.82)	48.0
R_2^{opt}	0.16 (3.89)	0.00 (0.24)	0.01 (0.36)	-0.01 (-0.47)	0.06 (1.98)	0.10 (1.16)	16.3
Panel C: RP-PCA Five Factors							
	α	β_1	β_2	β_3	β_4	β_5	Adj R^2 (%)
R_2^d	0.27 (6.38)	-0.04 (-2.81)	-0.05 (-2.46)	-0.02 (-0.36)	0.09 (1.12)	0.00 (0.02)	44.0
R_2^{opt}	0.22 (6.43)	-0.01 (-0.50)	-0.04 (-2.37)	0.01 (0.24)	0.06 (1.03)	-0.02 (-0.29)	7.83

gency portfolio, as the alpha estimates are approximately 0.21% and 0.16%, respectively, which are highly statistically significant in both regressions. While three IPCA factors are important in explaining excess returns on the deep factor, only one appears to be important in spanning the deep tangency portfolio. The adjusted R^2 s are approximately 48.0% and 16.3%, respectively, representing improvements over those obtained with the Fama-French five-factor model.

Similarly, we also observe in Panel C that the RP-PCA's five factors, constructed from 60 tercile portfolios, do not fully span the excess returns on the deep factor and

the deep tangency portfolio. The alpha estimates are about 0.27% ($t = 6.38$) and 0.22% ($t = 6.43$) and the adjusted R^2 s are 44.0% and 7.83%, respectively, in the two spanning regressions. The marginal improvement of our deep tangency portfolio over two PCA-based portfolios could be attributed to the nonlinearity in our deep learning framework, which is useful for return forecasting.

4.4 Interpreting Deep Characteristics

A key feature of our deep parametric portfolio framework is its balance between flexibility afforded by deep learning and transparency. This balance is achieved by anchoring the deep factor in an economically motivated Sharpe ratio loss function and structuring it as a long-short portfolio, thereby allowing us to unpack how firm characteristics drive the performance of the deep tangency portfolio. To deepen understanding of its economic mechanism, we focus on two critical dimensions: the importance of individual characteristics in shaping the deep factor and the incremental value of combining information from bonds, equities, and options.

The nonlinear *tanh* activation function in our neural network transforms those initial 132 high-dimensional characteristics into a single deep characteristic $Z_{t-1}^{(L)}$ (defined in Section 2.2), whose functional form is complex but interpretable via sensitivity analysis. Following [Chen et al. \(2024\)](#), we measure each characteristic's importance by its sensitivity to $Z_{t-1}^{(L)}$, which is quantified as the average absolute gradient of the deep characteristic with respect to the raw characteristic. Formally, this sensitivity is defined as:

$$\text{Sensitivity}(z_j) = \frac{1}{C} \sum_{i=1}^N \sum_{t=1}^T \left| \frac{\partial Z_{i,t-1}^{(L)}}{\partial z_j} \right|, \quad (23)$$

where $j = 1, \dots, 132$ (one for each characteristic), C normalizes sensitivities to an interpretable scale, N denotes the number of corporate bonds in each cross section, and T covers the out-of-sample period. This metric captures the extent to which each raw characteristic influences the deep characteristic and, by extension, the portfolio

weights of the deep factor.

Figure 4 plots these sensitivities, with bond, equity, and option characteristics distinguished by blue, yellow, and red bars, respectively. Our sensitivity analysis reveals two primary patterns: first, the deep characteristic is driven by a dense set of variables rather than a sparse selection. This aligns with Kozak et al. (2020) and Kozak and Nagel (2024), who argue that spanning the efficient frontier of asset returns requires leveraging a broad set of characteristics rather than sparse selections. Second, bond-related variables dominate the sensitivity rankings, with the top five including short-term reversal (STR), duration (DUR), time-to-maturity (T2M), term factor beta (TERM.BETA), and 12-month momentum (MOM12), which are economically salient corporate bond attributes and return signals. The emphasis on short-term reversals and duration-related factors highlights the deep factor’s outperformance, driven by the systematic capture of liquidity premiums and term risk compensation, two key return drivers in the fragmented, often illiquid OTC corporate bond market. Among non-bond characteristics, equity short-term reversal (MOM1M, 7th rank) and volatility of ATM option-implied volatility (IVVOL, 13th rank) stand out, highlighting meaningful cross-market information spillovers.

To further examine the importance of different types of characteristics, we reconstruct the softmax ranking-based deep tangency portfolio using equity and bond characteristics with option-related variables removed, and using bond or equity characteristics alone. Table 5 presents the Sharpe ratios of the deep tangency portfolios based on different characteristics, with the equal-weighted market factor as the benchmark. As shown above, when utilizing all characteristics, the 2-layer neural network performs the best, and the deep tangency portfolio achieves an OOS annualized Sharpe ratio of 2.13. However, excluding the option-related variables results in an OOS Sharpe ratio of about 1 for the deep tangency portfolio. We note that it now appears the 1-layer neural network is working well. This suggests that option-related variables contain

Figure 4: Variable Importance: Average Absolute Gradients

This figure presents the average absolute gradients of all characteristics using Equation (23) for the out-of-sample period. The blue bars represent bond characteristics, the yellow bars represent equity characteristics, and the red bars represent option characteristics. The OOS period ranges from July 2014 to December 2020.

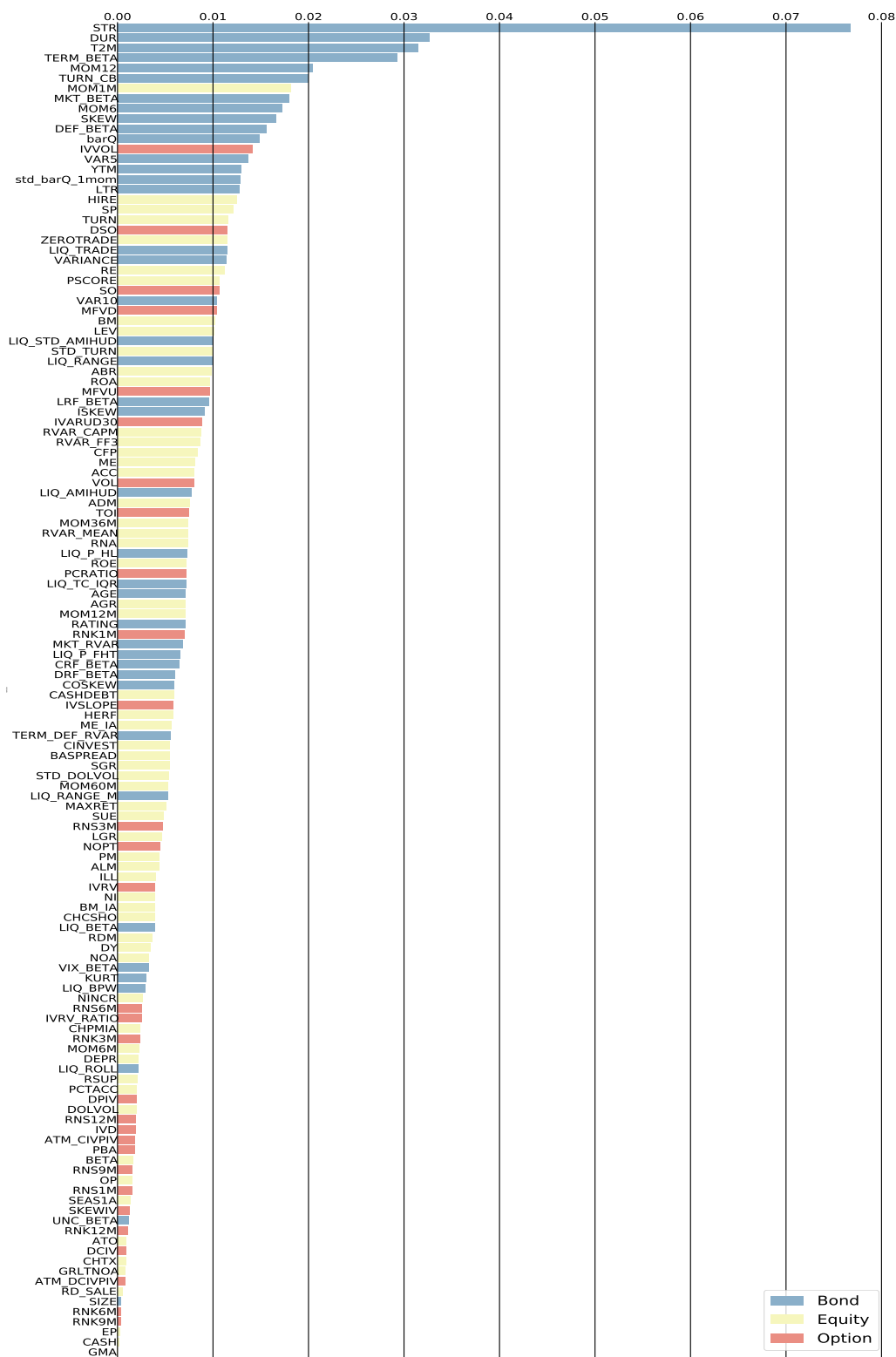


Table 5: Importance of Characteristics

This table presents the out-of-sample annualized Sharpe ratios obtained from different types of characteristics. We consider four sets of characteristics: all 132 characteristics, 102 bond and equity characteristics, 41 bond characteristics, and 61 equity characteristics. The test of the Sharpe ratio improvement of the deep tangency portfolio over the market factor is based on [Barillas and Shanken \(2017\)](#). ***, **, and * denote the levels of significance of 1%, 5%, and 10%, respectively. L_i represents a neural network with i layers. We use the equal-weighted market factor as the benchmark and employ the softmax ranking. The sample from July 2004 to June 2014 is used for model training and validation, while the sample from July 2014 to December 2020 is used for out-of-sample testing.

	MKT	Bond+Equity +Option	Bond+Equity	Bond	Equity
L_1	0.85	0.80	1.04***	0.76	0.83
L_2	0.85	2.13***	1.01***	0.70	0.57
L_3	0.85	0.95**	0.83	0.24	0.59

valuable information on future corporate bond returns. What is even worse is that when we use bond or equity characteristics alone, the performance of the deep tangency portfolio deteriorates even further.

Our results confirm that bond, equity, and option characteristics provide complementary return signals, which are essential for spanning the efficient frontier. The significant performance degradation when excluding any single domain validates the necessity of a multi-market information set for robust portfolio construction in the corporate bond market. This finding is, in fact, in stark contrast to previous studies that argue that those characteristics that predict equity returns do not necessarily forecast corporate bond returns (see, e.g., [Chordia et al., 2017](#); [Bali et al., 2021](#)).

Our findings provide new empirical evidence of the integration of bond, equity, and options markets through portfolio efficiency, rather than mere cross-market correlations. Unlike prior studies that document bond-equity linkages (e.g., [Schaefer and Strebulaev, 2008](#)), we demonstrate that combining characteristics from all three markets is crucial for improving the performance of corporate bond portfolios. This result aligns with [Dickerson et al. \(2023\)](#) and [Kelly et al. \(2023\)](#), who find equity characteristics add value to bond return prediction, and extends their insights by demonstrating

that option characteristics further enhance portfolio efficiency.

4.5 How Important Are the Economic Structure and Nonlinearity?

The deep factor in our parametric portfolio framework arises from a nonlinear transformation of raw characteristics, integrated with benchmark strategies to optimize the Sharpe ratio of the deep tangency portfolio. We assess the incremental benefit of this economic structure by comparing it with an unrestricted machine learning baseline, in which portfolio weights depend flexibly on characteristics.

To benchmark our architecture against existing standards, we construct comparative portfolios using return forecasts from a suite of machine learning models, ranging from Lasso and Ridge to standard multi-layer neural networks. At each time t , we forecast corporate bond returns using 132 firm-level characteristics across bonds, equities, and options via the selected ML models:

$$W_{t-1} \equiv E_{t-1}[r_t] = g(z_{t-1}, b), \quad (24)$$

where $g(\cdot)$ is the function learned by the ML model, and b aggregates model parameters. We consider two objective functions, as in [Bryzgalova et al. \(2025\)](#): maximizing the portfolio Sharpe ratio and minimizing the mean squared forecasting error (MSE). Portfolios are constructed via linear weighting or cross-sectional sorting. For linear weighting, W_{t-1} is simply taken as the portfolio weights, and for cross-sectional sorting, we long the top 10% (decile), 20% (quintile), or 30% (tercile) of bonds and short the corresponding bottom quantiles based on W_{t-1} resulting from the MSE objective. The sample from July 2004 to June 2014 is used for training and validation, while July 2014 to December 2020 serves as the OOS testing period. Detailed implementation steps are provided in Section [V](#) of the Online Appendix.

Panel A of Table [6](#) reports the OOS annualized Sharpe ratios of ML forecast-implied long-short portfolios. A clear observation emerges: none of the ML portfolios matches the efficiency of our deep tangency portfolio. While standard neural networks pro-

Table 6: ML Portfolios and Linear Activation

Panel A reports the out-of-sample annualized Sharpe ratios of various ML portfolios from neural networks with 1-3 layers (NN1, NN2, and NN3), Lasso, Ridge, and PCA. Two objective functions are considered: maximizing the portfolio's Sharpe ratio (Max_SR) and minimizing the mean squared forecasting error (Min_MSE). Portfolios are constructed via linear weighting or cross-sectional sorting. For linear weighting, W_{t-1} is simply set equal to the portfolio weights, and for cross-sectional sorting, we construct long-short portfolios based on W_{t-1} resulting from the MSE objective. Panel B presents the out-of-sample annualized Sharpe ratios of the softmax-ranking-based deep tangency portfolios, taking the equal-weighted market factor as the benchmark and replacing the nonlinear activation function with a linear one in the deep learning model. L_i represents a neural network with i layers. The sample from July 2004 to June 2014 is for model training and validation, and that from July 2014 to December 2020 is for out-of-sample testing.

Panel A: ML Portfolios					
Models	Max_SR	Min_MSE	10%	20%	30%
NN1	1.23	0.85	1.52	1.44	1.40
NN2	1.08	0.83	1.06	1.14	1.19
NN3	1.28	0.84	0.93	0.95	0.96
LASSO	0.84	1.06	0.86	0.81	0.85
Ridge	1.54	1.32	1.01	1.12	1.10
PCA	0.72	0.72	0.56	0.61	0.64
Panel B: Linear Activation					
	L_1	L_2	L_3		
SR	0.79	0.53	0.84		

duce predictive return forecasts, they lack the specific asset pricing objective of our framework, resulting in obviously lower out-of-sample portfolio efficiency. Its Sharpe ratio (1.52 from the NN1 decile portfolio) remains well below 2.13 achieved by our 2-layer deep tangency portfolio. Ridge-based portfolios perform reasonably well, but their Sharpe ratios are even lower (the largest is 1.54 under the SR objective with linear weighting). These results highlight the importance of our economic structure, which is explicitly designed to enhance the benchmark portfolio's performance.

Furthermore, to isolate the role of nonlinearity, we replace the *tanh* activation function in our deep learning framework with a linear function. Panel B of Table 6 presents results and reveals two critical findings: first, linear activation requires deeper networks (3 layers) to complement the benchmark, indicating its inefficiency in capturing

cross-market spillovers; second, even with the deeper network, the OOS Sharpe ratio of the linear-activation portfolio remains far below that of the nonlinear-activation one (0.84 for 3 layers vs. 2.13 for 2 layers). This highlights the importance of nonlinear transformations for uncovering residual risk-return trade-offs that linear models overlook.

In summary, two key elements drive the success of our framework: the nonlinear transformations, which capture complex relations among characteristics for return forecasting, and the economically motivated Sharpe ratio objective with an economic structure, which combines the benchmark factors and a nonlinear long-short deep factor. Together, these innovations enable the deep factor to enhance the performance of the benchmark strategies. These findings align with prior literature emphasizing the effects of nonlinearity in asset return predictability (see, e.g., [Freyberger et al., 2020](#); [Gu et al., 2020](#)) and the value of economic constraints in portfolio construction (see, e.g., [Jensen et al., 2026](#); [Avramov et al., 2023](#); [Liu et al., 2024](#)).

5 Conclusion

While the tangency portfolio remains a theoretical cornerstone of finance, its practical implementation in high-dimensional settings is notoriously difficult. Traditional factor models, limited to a narrow set of characteristics, fail to span the full efficient frontier because evidence for sparsity in asset pricing and portfolio construction is lacking ([Kozak et al., 2020](#); [Kozak and Nagel, 2024](#)). Our study addresses this issue by proposing a parametric deep learning framework that aims to improve the performance of well-established benchmark strategies. By formalizing benchmark factors as an economic prior, our framework imposes economic regularization and mitigates the challenge of low signal-to-noise ratios in financial data. Furthermore, the direct estimation of portfolio weights in our framework sidesteps the compounded errors inherent to traditional plug-in methods.

Guided by an economically motivated objective of maximizing the Sharpe ratio, our approach constructs an endogenous deep factor from high-dimensional characteristics, which is then combined with benchmark factors to form a deep tangency portfolio. The deep factor enhances the benchmark by incorporating additional signals from multi-market cross-sectional characteristics and imposing a long-short structure. It exhibits a low or even negative correlation with the benchmark, providing a natural hedge against benchmark risks.

Our application to the U.S. corporate bond market demonstrates that the resulting deep tangency portfolio, created by complementing the benchmark equal-weighted (value-weighted) market factor with the deep factor, delivers an annualized Sharpe ratio of 2.13 (2.21), substantially outperforming optimal portfolios constructed from commonly used observable or latent factors, such as the Fama-French five factors that include three bond factors and two equity factors, and the five latent factors derived from IPCA or RP-PCA. Our empirical analysis reveals the importance of various firm characteristics in portfolio construction. Notably, the deep factor employs a broad set of characteristics, providing further evidence against sparsity. These results suggest that spanning the corporate bond efficient frontier requires a large number of multi-asset characteristics, which contrasts with studies questioning the relevance of equity characteristics to bond returns, but aligns with evidence of bond-equity market integration.

Finally, our evidence should be interpreted in light of the sample period and empirical setting. The OOS evaluation ends in December 2020 and therefore does not cover the subsequent post-2021 rising-rate cycle; assessing the framework in a sustained rising-rate environment is a natural direction for future research. More broadly, although our empirical application focuses on U.S. corporate bonds, the benchmark-augmentation logic of the framework can be adapted to other asset classes, with the network and other hyperparameters selected and validated in each setting.

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Online Appendix for “Deep Tangency Portfolio”

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Summary of Contents

- Section **I** presents detailed definitions of bond, equity, and option characteristics used in our empirical studies.
- Section **II** presents the implementation details of our deep learning algorithm and some robustness checks.
- Section **III** presents the results from two alternative sample split schemes.
- Section **IV** presents the performance of the deep tangency portfolio accounting for transaction costs.
- Section **V** presents the implementation details of machine learning (ML) algorithms.

I Bond, Equity, and Option Characteristics

Table A1: Description of 41 Bond Characteristics

Characteristics	Description
AGE	Time since issuance in years
RATING	Bond credit rating
T2M	The number of years to maturity
SIZE	Amount outstanding
DUR	Bond duration
VAR5	Value-at-risk 5% over past 3 years
VAR10	Value-at-risk 10% over past 3 years
LIQ_BPW	Liquidity measure of transitory price movements.
LIQ_ROLL	Roll's liquidity
LIQ_P_HL	High-low spread estimator
LIQ_P_FHT	Modified illiquidity measure based on zero returns
LIQ_AMIHUD	Amihud liquidity
LIQ_STD_AMIHUD	Standard deviation of Amihud daily liquidity
LIQ_TC_IQR	Interquartile range
MKT_BETA	Market beta
DEF_BETA	DEF factor beta
TERM_BETA	TERM factor beta
LIQ_BETA	Liquidity beta of bond illiquidity factor
DRF_BETA	Downside risk beta controlling bond market factor
CRF_BETA	Credit risk beta controlling bond market factor
LRF_BETA	Liquidity risk beta controlling bond market factor
VIX_BETA	VIX index beta
UNC_BETA	Macroeconomic Uncertainty Beta
STR	Short-term reversal t-1
VARIANCE	Variance of raw returns
SKEW	Skewness of raw returns
KURT	Kurtosis of raw returns
COSKEW	Systematic skewness with bond market
ISKEW	Idiosyncratic skewness
LIQ_RANGE	Simple high-low spread
LIQ_TRADE	Number of trades
MKT_RVAR	Market residual variance
TERM_DEF_RVAR	TERM DEF residual variance
TURN_CB	Bond Turnover
YTM	Yield-to-maturity
MOM6	Momentum from t-2 to t-6
MOM12	Momentum from t-7 to t-12
LTR	Long-term reversal from t-13 to t-48
BARQ	Average daily dollar volume in the 1-month period
STD_BARQ_1MOM	Standard deviation of dollar volume in the 1-month period
LIQ_RANGE_M	Simple high-low spread

Table A2: Description of 61 Equity Characteristics

Characteristics	Description
ABR	Abnormal returns around earnings announcement
ACC	Operating Accruals
ADM	Advertising Expense-to-market
AGR	Asset growth
ALM	Quarterly Asset Liquidity
ATO	Asset Turnover
BASPREAD	Bid-ask spread (3 months)
BETA	Beta (3 months)
BM	Book-to-market equity
BM.IA	Industry-adjusted book to market
CASH	Cash holdings
CASHDEBT	Cash to debt
CFP	Cashflow-to-price
CHCSHO	Change in shares outstanding
CHPMIA	Industry-adjusted change in profit margin
CHTX	Change in tax expense
CINVEST	Corporate investment
DEPR	Depreciation / PP&E
DOLVOL	Dollar trading volume
DY	Dividend yield
EP	Earnings-to-price
GMA	Gross profitability
GRLTNOA	Growth in long-term net operating assets
HERF	Industry sales concentration
HIRE	Employee growth rate
ILL	Illiquidity rolling (3 months)
LEV	Leverage
LGR	Growth in long-term debt
MAXRET	Maximum daily returns (3 months)
ME	Market equity
ME.IA	Industry-adjusted size
MOM12M	Cumulative Returns in the past (2-12) months
MOM1M	Previous month return
MOM36M	Cumulative Returns in the past (13-35) months
MOM60M	Cumulative Returns in the past (13-60) months
MOM6M	Cumulative Returns in the past (2-6) months
NI	Net Equity Issue
NINCR	Number of earnings increases
NOA	Net Operating Assets
OP	Operating profitability
PCTACC	Percent operating accruals
PM	Profit margin
PS	Performance Score
RD_SALE	R&D to sales
RDM	R&D Expense-to-market
RE	Revisions in analysts' earnings forecasts
RNA	Return on Net Operating Assets
ROA	Return on Assets
ROE	Return on Equity
RSUP	Revenue surprise
RVAR_CAPM	Residual variance - CAPM (3 months)
RVAR_FF3	Res. var. - Fama-French 3 factors (3 months)
SVAR	Return variance (3 months)

Description of 61 Equity Characteristics (continued)

Characteristics	Description
SEAS1A	1-Year Seasonality
SGR	Sales growth
SP	Sales-to-price
STD_DOLVOL	Std of dollar trading volume (3 months)
STD_TURN	Std. of Share turnover (3 months)
SUE	Unexpected quarterly earnings
TURN	Shares turnover
ZEROTRADE	Number of zero-trading days (3 months)

Table A3: Description of 30 Equity Option Characteristics

Characteristics	Description
IVSLOPE	Implied Volatility Slope
IVVOL	Volatility of atm implied volatility
IVRV	Implied and historical volatility spread
IVRV_RATIO	Ratio of implied to historical volatility
ATM_CIVPIV	Implied volatility spread
SKEWIV	Implied volatility skew
IVD	Implied volatility duration
DCIV	Change of implied volatility of atm call
DPIV	Change of implied volatility of atm put
ATM-DCIVPIV	Change of implied volatility spread
NOPT	Number of traded options
SO	Stock-option volume ratio
DSO	Stock-option dollar volume ratio
VOL	Option Trading Volume
PCRATIO	Put-call ratio
PBA	Proportional bid-ask spread
TOI	Total open interest
MFVU	Option-implied upside semivariance
MFVD	Option-implied downside semivariance
RNS1M	1-month risk-neutral skewness
RNK1M	1-month risk-neutral kurtosis
IVARUD30	Option-implied variance asymmetry
RNS3M	3-month risk-neutral skewness
RNK3M	3-month risk-neutral kurtosis
RNS6M	6-month risk-neutral skewness
RNK6M	6-month risk-neutral kurtosis
RNS9M	9-month risk-neutral skewness
RNK9M	9-month risk-neutral kurtosis
RNS12M	12-month risk-neutral skewness
RNK12M	12-month risk-neutral kurtosis

II Implementation Details

II.1 Data Cleaning

As shown in Table 1, we provide the summary statistics for all our Bond-Month observations (before standardization and imputation). Since the raw bond return data began in July 2002 and at least 1 year of data is required to initialize risk characteristics, we start with data from July 2004. After standardizing the raw characteristics, we filter out bonds with a maturity of less than 2 years and impute missing values with 0. The equity and option characteristics are merged into bond data by the “PERMNO”. If the firm has issued multiple stocks simultaneously, we keep only the earliest-issued stock and merge it.

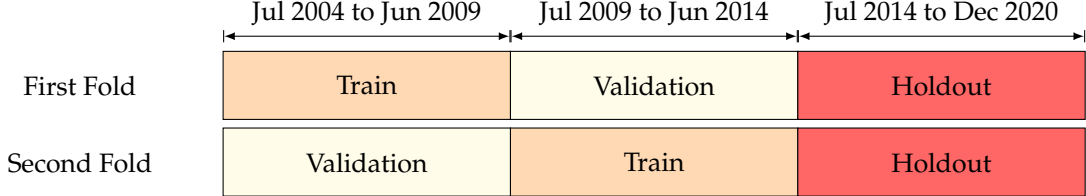
II.2 Model Training

We divide the TRACE dataset into two parts: we train on samples from July 2004 to June 2014 and test on samples from July 2014 to December 2020. We train the neural network’s structure and parameters during the in-sample period and determine the portfolio’s weights based on the in-sample factor statistics. To avoid the effect of outliers during training, we also winsorize the in-sample bond returns within the $ret_{low,t}$ and $ret_{up,t}$ bounds at each month t , where $ret_{low,t}$ and $ret_{up,t}$ are the cross-sectional 5% and 95% quantiles of month t , respectively. Each month, bond returns that are lower/higher than the monthly low/up quantiles of cross-sectional returns will be revised to $ret_{low,t}$ and $ret_{up,t}$, respectively. It is important to emphasize that winsorization is only applied to in-sample data. All out-of-sample (OOS) results are tested on NON-Winsorized data, which may contain potential extreme values.

We use both equal- and value-weighted market factors as our benchmarks. For the IPCA/RP-PCA model, we train the PCA structure using balanced individual bond returns from July 2004 to June 2014, and compute the factor values for both the in-

Figure A1: Two-Fold Cross Validation

This figure illustrates the deterministic two-fold cross-validation scheme used to select the tuning parameters. Specifically, we determine the tuning parameters using the sample from July 2004 to June 2014. The deterministic design divides the sample into two consecutive fold samples. We train our neural network separately on one, then calculate the fit results with different tuning parameters on the other. We calculate the average out-of-sample loss and select the parameter pair with the best performance.



sample and OOS periods using this trained structure. Similarly, our neural network takes the individual bond data from July 2004 to June 2014 as a training set for our neural network structure, and we determine the tuning parameters through a two-fold cross-validation process, as shown in Figure A1, for which we will provide more details in Appendix II.3.

In ranking the bond returns, we choose $a_1 = 50$ and $a_2 = 8$ in Equation (16) such that at each time, approximately 50 % to 70% of assets are in the middle rank and have negligible weights, similar to the traditional sorting procedure (see Figure A2).¹³

After training, we use the fixed network structure and input the pairwise characteristics to generate in-sample and OOS estimates. All results in Section II are computed using PyTorch 1.10.2 and parallelized across a server with 96 Intel(R) Xeon (R) Gold 6230 @ 2.10GHz CPUs and 314 GB of RAM.

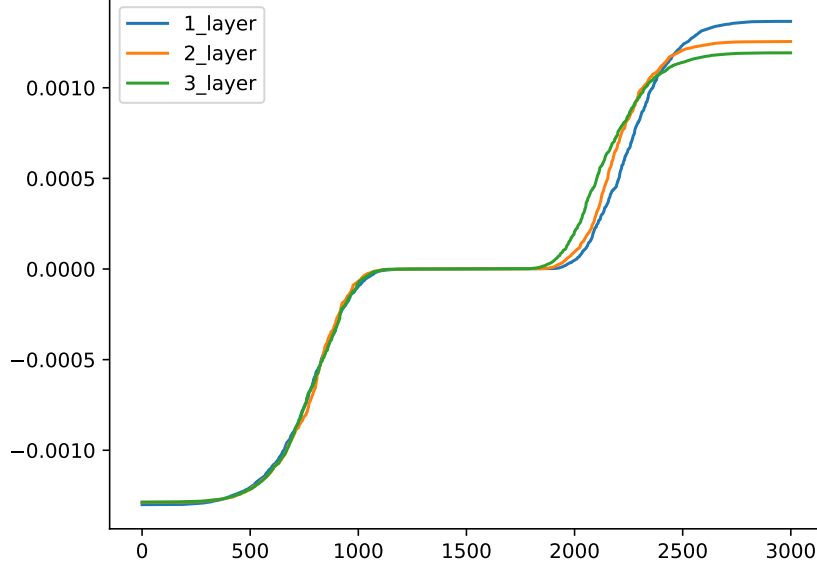
II.3 Tuning Parameters Selection

In this subsection, we outline our procedure for selecting tuning parameters. To determine the optimal tuning parameters for our network, we employ a two-fold cross-validation approach (as shown in Figure A1) as follows. We first divide the sample

¹³We select a_1 and a_2 sequentially rather than through a full two-dimensional grid because both parameters primarily govern the concentration of the long and short tails. Moderate changes around the selected values produce similar S-shaped weighting profiles and do not change the qualitative ranking of the network architectures. The pair $a_1 = 50$ and $a_2 = 8$ minimizes the average validation loss among the values considered.

Figure A2: Softmax Ranking ($a_1 = 50$ and $a_2 = 8$)

This figure plots the individual corporate bonds' weights generated by the softmax rankings of deep characteristics for different layers.



period from July 2004 to June 2014 into two consecutive fold samples of equal length. We train our neural network on one fold and evaluate the fitted results using different tuning parameters on the other fold. We compute the average loss over the validation samples and select the parameter pairs that yield the smallest loss. Finally, we retrain the model using the chosen tuning parameters. This approach ensures that the data used is completely in-sample, eliminating any look-ahead bias that may affect our OOS trading results discussed in the main text. When implementing, we start with a reasonable set of initial tuning parameters and test additional points adjacent to these sets. We evaluate a total of 120 combinations of tuning parameters, which are shown in Table A4.

III Alternative Sample Splits

III.1 Reversed Sample

To ensure robustness in OOS evaluation, we reverse the training, validation, and forecasting periods in this subsection. In other words, we use the end periods of the

Table A4: Tuning Parameter Selection in the Empirical Analysis

This table presents the network tuning parameters used in our empirical analysis.

Notation	Tuning Parameters	Candidates	Chosen
a_1	Value in equation (16)	50	50
a_2	Value in equation (16)	5,8,10	8
HDN	Number of nodes in the hidden layer	66	66
BTCH	Batch size, in months	10,30	10
LR	Learning rate	1e-3, 5*1e-3, 1e-2, 5*1e-2, 1e-1	1e-3
DR	Dropout rate	0.05, 0.1, 0.2, 0.3	0.1
EPCH	Number of optimization epochs	400	400
OPT	Optimization method	Adam	Adam

original sample as training and validation sets and the beginning periods as the OOS evaluation periods.

We use the same approach as in Section 2.2 to build the neural network. Panel A in Table A5 shows that, 1) the deep tangency portfolios have significantly higher OOS Sharpe ratios than the market portfolio; 2) the 2-layer deep tangency portfolios achieve the highest OOS Sharpe ratio in most cases, both of which are consistent with our previous findings.

III.2 Sample Randomly Split

To further ensure robustness in OOS evaluation, we follow Fama and French (2018) to randomly split the dataset into three parts sequentially. Specifically, in every quarter (three months), the data is randomly assigned to one of three groups. This process continues until each month's data is assigned to a group. Figure A3 illustrates how we divide the dataset into three groups, represented by different colors. We then employ the same approach as in Section 2.2 to construct the neural network and implement a two-fold cross-validation scheme: for each test group, the other two groups are used for training and validation. Panel B in Table A5 shows the OOS performance for all three groups. We find that: 1) the deep tangent portfolios have significantly higher OOS Sharpe ratios than the market portfolio; 2) the layer-2 deep tangent portfolios achieve the highest OOS Sharpe ratio in most cases. These results are consistent with our previous findings.

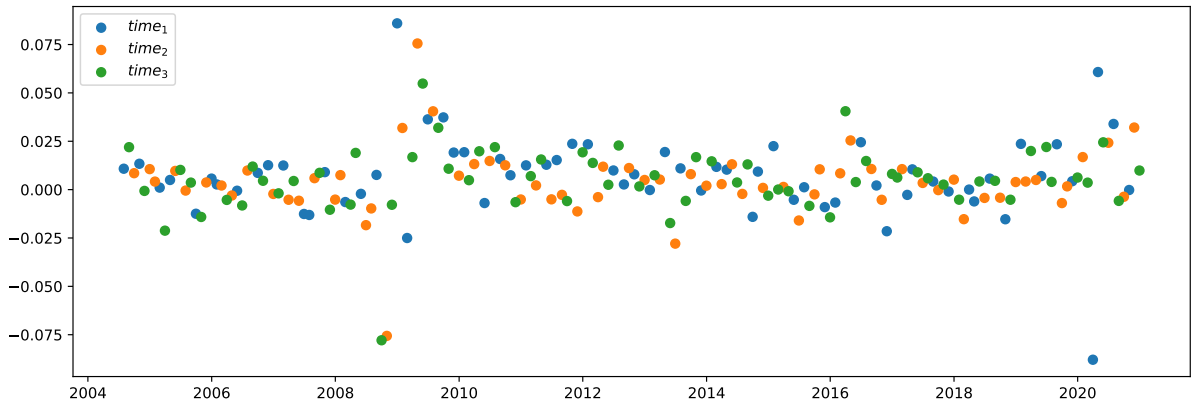
Table A5: Deep Tangency Portfolios in Two Alternative Sample Split Schemes

This table reports the out-of-sample annualized Sharpe ratios of the deep tangency portfolios using multiple-layer deep factors from two alternative sample split schemes. L_i represents a neural network with i layers. We respectively describe two panels as follows: (i) Panel A presents the performance of the reversed sample. We use samples from January 2011 to December 2020 for model training and validation, and from July 2004 to December 2010 for out-of-sample testing; and (ii) Panel B displays the performance for the randomly split sample. We follow [Fama and French \(2018\)](#) to randomly split the dataset into three parts sequentially. Specifically, in every quarter (three months), the data is randomly assigned to one of three groups. We employ the same approach as in Section 2.2 to construct the neural network and implement a two-fold cross-validation scheme: for each test group, the other two groups are used for training and validation. For example, the “ $time_1$ ” column displays the out-of-sample performance when groups 2 and 3 are used for training and validation, respectively, and group 1 is used for the out-of-sample evaluation. The “ all ” column displays the out-of-sample Sharpe ratios for all three groups. Additionally, in both panels, “MKT” represents the market factor, which is compared with the other three multiple-layer deep tangency portfolios. Moreover, in Panels A, the test of the Sharpe ratio improvement by including an additional deep factor is based on [Barillas and Shanken \(2017\)](#). ***, **, and * denote the levels of significance of 1%, 5%, and 10%, respectively.

	Equal-Weighted				Value-Weighted			
Panel A: Reversed Sample								
	MKT	L_1	L_2	L_3	MKT	L_1	L_2	L_3
SR	0.87	1.41***	1.59***	1.26***	0.78	1.27***	1.50***	1.39***
Panel B: Sample Randomly Split								
	$time_1$	$time_2$	$time_3$	all	$time_1$	$time_2$	$time_3$	all
MKT	1.14	0.73	1.06	0.98	1.11	0.74	0.91	0.93
L_1	2.82	1.76	2.47	2.19	3.30	1.97	0.99	1.70
L_2	3.07	2.83	2.17	2.65	2.85	2.37	2.08	2.41
L_3	2.47	1.93	2.21	2.18	2.22	2.26	2.47	2.31

Figure A3: Bond Market Return for the Sample Randomly Split

This figure plots the bond market return from July 2004 to December 2020. Different colors represent three randomly split groups, denoted by $time_1$, $time_2$, and $time_3$, respectively.



IV Deep Tangency Portfolios Accounting for Transaction Costs

Following [Kelly, Palhares, and Pruitt \(2023\)](#), we incorporate transaction costs based on turnover. After the weights of a deep factor are generated by high-dimensional characteristics as in Equation (14), we replace the raw weights $w_d(z_{t-1})$ by the “slowed down” weights $\tilde{w}_d(z_{t-1})$ according to exponential smoothing as follows:

$$\tilde{w}_d(z_{t-1}) = (1 - \gamma)w_d(z_{t-1}) + \gamma\tilde{w}_d(z_{t-2}), \quad (25)$$

where γ is a tuning hyperparameter that can control the degree of smoothing, and we initialize with $\tilde{w}_d(z_{-1}) = w_d(z_0)$. Subsequently, we calculate the net deep factor returns, $\tilde{R}_{d,t}$, by subtracting the costs from gross portfolio returns in Equation (18),

$$\begin{aligned} \tilde{R}_{d,t} &= R_{d,t} - R_{TC,t} \\ &= \tilde{w}_d(z_{t-1})r_t - \lambda \times |\tilde{w}_d(z_{t-1}) - \tilde{w}_d(z_{t-2})|, \end{aligned} \quad (26)$$

where $|\tilde{w}_d(z_{t-1}) - \tilde{w}_d(z_{t-2})|$ is the portfolio turnover as the ℓ_1 distance between consecutive smoothed weight vectors, and λ represents the constant transaction costs. To be conservative, we apply one-way costs of two representative levels ($\lambda = 15$ and 20 bps) to all trades. Finally, we implement the same approach as in Section 2.2 and add a range constraint for the tangency portfolio weights to construct the deep tangency portfolio.¹⁴

Table A6 reports Sharpe ratios of the resulting deep tangency portfolios across different network layers, smoothing parameters, and transaction cost levels. A central finding is that incorporating transaction costs does not alter the main conclusions of our paper. Across both equal-weighted and value-weighted specifications, the relative ranking of models remains stable. In particular, the 2-layer network consistently delivers the highest Sharpe ratios, even after accounting for turnover-based trading

¹⁴The case including transaction costs is more complicated. To avoid extreme short positions (e.g., [Ferreira and Santa-Clara, 2011](#); [Thornton and Valente, 2012](#)) after including transaction costs, we restrict the tangency portfolio weights to be between -2 and 3.

Table A6: Deep Tangency Portfolio Accounting for Transaction Cost

This table presents the out-of-sample annualized Sharpe ratios of the deep tangency portfolios constructed using multiple-layer deep factors accounting for transaction costs. L_i represents a neural network with i layer(s), and γ is the parameter for turnover levels. “MKT” is the market factor, which serves as the benchmark for comparison with the other three multiple-layer deep tangency portfolios. The test of the Sharpe ratio improvement by including an additional deep factor is based on [Barillas and Shanken \(2017\)](#). ***, **, and * denote the level of significance of 1%, 5%, and 10%, respectively. We take the sample from July 2004 to June 2014 for model training and validation, and the sample from July 2014 to December 2020 for out-of-sample testing.

γ	Equal-Weighted				Value-Weighted			
	MKT	L_1	L_2	L_3	MKT	L_1	L_2	L_3
Panel A: 15 Basis Points								
0.0	0.85	0.15	1.50***	0.28	0.90	0.12	1.91***	0.40
0.1	0.85	0.44	1.34***	0.38	0.90	0.34	1.62***	0.41
0.3	0.85	0.77	1.14***	0.86	0.90	0.85	1.62***	0.90
0.5	0.85	0.74	1.06***	0.71	0.90	0.62	1.07***	0.78
0.7	0.85	0.89	1.18***	0.98***	0.90	0.91	1.15***	0.95
0.9	0.85	0.85	0.93**	0.88	0.90	0.93	0.98**	0.93
Panel B: 20 Basis Points								
0.0	0.85	0.14	1.52***	1.01***	0.90	0.24	1.87***	1.06***
0.1	0.85	0.75	1.54***	0.31	0.90	0.27	1.92***	0.28
0.3	0.85	0.38	1.17***	0.97***	0.90	0.51	1.23***	1.00**
0.5	0.85	0.83	1.08***	0.90	0.90	0.99**	1.18***	0.93
0.7	0.85	0.77	1.00***	0.86	0.90	0.88	0.99**	0.89
0.9	0.85	0.85	0.92*	0.87	0.90	0.93	0.97*	0.90

costs and weight smoothing.

More broadly, the results indicate that the performance of the deep factors is robust to realistic implementation frictions. While higher transaction costs and stronger smoothing naturally attenuate returns, they do not eliminate the economic advantage of deeper architectures. The persistence of the 2-layer model’s outperformance suggests that its superior risk-adjusted returns are not driven by excessive trading, but by more informative and stable cross-sectional signals. Taken together, these findings reinforce the empirical relevance of deep tangency portfolios when transaction costs are explicitly accounted for.

V The Implementations of Standard Machine Learning Methods

In Section 4.5, we present a forecast-implied long-short portfolio based on predicted bond returns from four ML methods: Lasso, Ridge, principal component anal-

ysis regression (PCA), and standard neural networks (NN) to predict the asset returns for comparison directly. Our implementation follows [Feng et al. \(2025\)](#).

PCA is a common dimension reduction technique used in empirical asset pricing to address the bias-variance trade-off. It involves using a low-dimensional linear transformation of the predictors to construct a predictive model. The procedure consists of two steps: the first combines predictors with a small set of linear combinations that best preserve the covariance structure, and the second involves using the first K components in multiple regressions. In our applications, K is set to 5 for PCA.

Lasso and ridge are linear regularized regressions used in ML finance to maintain interpretability. They add a penalty over least squares to preserve the predictors without transforming them. Lasso performs variable selection, while ridge shrinks the regression coefficients of useless predictors to small values. A tuning parameter controls the penalty weight, with a larger penalty weight imposing more shrinkage on the coefficients. A two-fold cross-validation is used to determine all tuning parameters.

For the standard neural network (NN), we use the same input characteristics and two-fold cross-validation scheme as in our deep learning network. We also add regularization penalty terms to the loss function to limit the complexity of the neural networks, stabilize estimates, and avoid overfitting.